Exercise J-M

1. For $k \in N$, let $\frac{1}{\alpha(\alpha+1)(\alpha+2).....(\alpha+20)} = \sum_{k=0}^{20} \frac{A_k}{\alpha+k}$, where $\alpha > 0$. Then the value of $100\left(\frac{A_{14}+A_{15}}{A_{13}}\right)^2$ is equal to_____. **[JEE(Main)July2021]**

माना $k \in N$ के लिए, $\frac{1}{\alpha(\alpha+1)(\alpha+2).....(\alpha+20)} = \sum_{k=0}^{20} \frac{A_k}{\alpha+k}$, $\alpha > 0$ है। तो $100\left(\frac{A_{14}+A_{15}}{A_{13}}\right)^2$ बराबर है _____ **[JEE(Main)July2021]**

Ans. 9

Sol. by partial fraction

for A_{14} put $\alpha = -14$ in $\frac{1}{\alpha(\alpha+1)(\alpha+2).....(\alpha+20)}$

$$A_{14} = \frac{1}{(-14)(-13).....(-1)(1)(2).....(6)}$$

$$A_{15} = \frac{1}{(-15)(-14).....(-1)(1)(2).....(5)}$$

$$A_{13} = \frac{1}{(-13)(-12).....(-1)(1)(2).....(7)}$$

$$100\left(\frac{A_{14}+A_{15}}{A_{13}}\right)^2 = 100\left\{\frac{\frac{-15+6}{(-15)(-14).....(-1)(1)(2).....(6)}}{\frac{1}{(-13)(-12).....(-1)(1)(2).....(7)}}\right\}^2$$

$$= 100\left(\frac{\frac{-9}{(-15)(-14)}}{\frac{1}{7}}\right)^2$$

$$= 100\left(\frac{-3}{10}\right)^2 = 9$$

2. Let a be a positive real number such that $\int_0^a e^{x-[x]}\,dx = 10e-9$ where [x] is the greatest integer less than or equal to x. Then a is equal to : **[JEE(Main)July2021]**

माना a एक धनात्मक वास्तविक संख्या है जिसके लिए $\int_0^a e^{x-[x]}\,dx = 10e-9$ है, जहाँ [x] महतम पूर्णांक $\le$ x है। तो a बराबर है:

[JEE(Main)July2021]

(A) $10 + \log_e(1+e)$ (B) $10 + \log_e 2$ (C) $10 - \log_e(1+e)$ (D) $10 + \log_e 3$

Ans. B

Sol. $\int_0^{[a]} e^{x-[x]}dx + \int_{[a]}^{a} e^{x-[x]}dx = 10e-9$

$[a]\int_0^1 e^{x-[x]}dx + \int_0^{\{a\}} e^{x-[x]}dx = 10e-9$

$[a]\int_0^1 e^{x}dx + \int_0^{\{a\}} e^{x}dx = 10e-9$

$[a](e-1) + e^{\{a\}} - 1 = 10e-9$

$[a] = 10$ $\qquad -[a] + e^{\{a\}} - 1 = -9$

$e^{\{a\}} = 2$

$\{a\} = \log_e 2$

So $a = [a] + \{a\}$

$= 10 + \log_e 2$

3. The value of the integral $\int_{-1}^{1} \log_e(\sqrt{1-x} + \sqrt{1+x})\,dx$ is equal to : **[JEE(Main)July2021]**

समाकलन $\int_{-1}^{1} \log_e(\sqrt{1-x} + \sqrt{1+x})\,dx$ का मान बराबर है : **[JEE(Main)July2021]**

(A) $\log_e 2 + \frac{\pi}{2} - 1$ (B) $2\log_e 2 + \frac{\pi}{2} - \frac{1}{2}$ (C) $\frac{1}{2}\log_e 2 + \frac{\pi}{4} - \frac{3}{2}$ (D) $2\log_e 2 + \frac{\pi}{4} - 1$

Ans. A

Sol. $I = \int_{-1}^{1} \ln(\sqrt{1-x} + \sqrt{1+x})\,dx$

$I = 2\int_0^1 \ln(\sqrt{1-x} + \sqrt{1+x})\,dx$

$= \int_0^1 \ln(2 + 2\sqrt{1-x^2})\,dx$

$= \int_0^1 \ln 2\,dx + \int_0^1 \ln(1+\sqrt{1-x^2})\,dx$

Integrate by parts

$$= \ln(2) + (x\ln(1+\sqrt{1-x^2}))_0^1 + \int_0^1 \frac{x^2\ dx}{(\sqrt{1-x^2})(1+\sqrt{1-x^2})}$$

Put $1-x^2 = t^2$

$$= \ln 2 + \int_0^1 \frac{(1-t^2)\ tdt}{t(1+t)\sqrt{1-t^2}}$$

$$= \ln 2 + \int_0^1 \sqrt{\frac{1-t}{1+t}}dt$$

$$= \ln 2 + \int_0^1 \frac{1-t}{\sqrt{1-t^2}}dt$$

$$= \ln 2 + (\sin^{-1}t)_0^1 + (\sqrt{1-t^2})_0^1$$

$$= \ln 2 + \left(\frac{\pi}{2} - 0\right) + (0-1)$$

$$= \log_e 2 + \frac{\pi}{2} - 1$$

4. If $[x]$ denotes the greatest integer less than or equal to x, then the value of the integral $\int_{-\pi/2}^{\pi/2} [[x] - \sin x]dx$ is equal to : **[JEE(Main)July2021]**

यदि $[x]$, महत्तम पूर्णांक $\le x$ है, समाकलन $\int_{-\pi/2}^{\pi/2} [[x] - \sin x]dx$ का मान बराबर है : **[JEE(Main)July2021]**

(A) 0 (B) π (C) $-\pi$ (D) 1

Ans. C

Sol. $I = \int_{-\pi/2}^{\pi/2} ([x] + [-\sin x])dx \{\because [x+I] = [x] + I\}$

Using property $\int_{-a}^{a} f(x)\,dx = \int_0^a f(x)\,dx + \int_0^a f(-x)\,dx$

$$I = \int_0^{\frac{\pi}{2}} ([x] + [-\sin x])dx + \int_0^{\frac{\pi}{2}} ([-x] + [-\sin x])dx$$

$$I = \int_0^{\pi/2} ([x] + [-x])\,dx + \int_0^{\pi/2} ([\sin x] + [-\sin x])\,dx$$

$$I = -\int_0^{\pi/2} dx - \int_0^{\pi/2} dx$$

$I = -\pi$

5. Let $g(t)=\int_{-\pi/2}^{\pi/2}\cos\left(\frac{\pi}{4}t+f(x)\right)dx$, where $f(x)=\log_e(x+\sqrt{x^2+1}), x\in R$. Then which one of the following is correct ? **[JEE(Main)July2021]**

माना $g(t)=\int_{-\pi/2}^{\pi/2}\cos\left(\frac{\pi}{4}t+f(x)\right)dx$, जबकि $f(x)=\log_e(x+\sqrt{x^2+1}), x\in R$, है। तो निम्न में से कौनसा सही है ? **[JEE(Main)July2021]**

(A) $g(1)=g(0)$ (B) $\sqrt{2}\,g(1)=g(0)$ (C) $g(1)+g(0)=0$ (D) $g(1)=\sqrt{2}\,g(0)$

Ans. B

Sol. $$g(1)=\int_{-\pi/2}^{\pi/2}\cos\left(\frac{\pi}{4}+f(x)\right)dx$$

$$g(1)=\int_{-\pi/2}^{\pi/2}\cos\frac{\pi}{4}\cos(f(x))dx-\int_{-\pi/2}^{\pi/2}\sin\frac{\pi}{4}\sin(f(x))dx$$

$$g(1)=\frac{1}{\sqrt{2}}\int_{-\pi/4}^{\pi/4}\cos(f(x))dx-\frac{1}{\sqrt{2}}\int_{-\pi/2}^{\pi/2}\sin\left(\log\left(n+\sqrt{x^2+1}\right)\right)dx$$

$\left\{\text{we know that } \sin\left(\log_e\left(4+\sqrt{x^2+1}\right)\right)\text{is a odd funtion}\right\}$

$$g(1)=\frac{1}{\sqrt{2}}\int_{-\pi/4}^{\pi/4}\cos(f(x))dx+0$$

$$\sqrt{2}g(1)=g(0)$$

6. If $f:R\to R$ is given by $f(x)=x+1$, then the value of $\lim_{n\to\infty}\frac{1}{n}\left[f(0)+f\left(\frac{5}{n}\right)+f\left(\frac{10}{n}\right)+....+f\left(\frac{5(n-1)}{n}\right)\right]$, is : **[JEE(Main)July2021]**

यदि $f:R\to R$, $f(x)=x+1$ द्वारा परिभाषित है, तो $\lim_{n\to\infty}\frac{1}{n}\left[f(0)+f\left(\frac{5}{n}\right)+f\left(\frac{10}{n}\right)+....+f\left(\frac{5(n-1)}{n}\right)\right]$ का मान है: **[JEE(Main)July2021]**

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$ (C) $\frac{7}{2}$ (D) $\frac{5}{2}$

Ans. C

Sol. $$\lim_{n\to\infty}\frac{1}{n}\left[f\left(\frac{0}{n}\right)+f\left(\frac{5}{n}\right)+f\left(\frac{10}{n}\right)+.....+f\left(\frac{5(n-1)}{n}\right)\right]$$

$$= \lim_{n\to\infty} \frac{1}{n} \sum_{r=0}^{n-1} f\left(\frac{5r}{n}\right)$$

$$= \int_0^1 f(5x)\,dx = \int_0^1 (5x+1)$$

$$= \left[5\frac{x^2}{2} + x\right]_0^1 = \frac{7}{2}$$

7. If $\int_0^{100\pi} \frac{\sin^2 x}{e^{\left(\frac{x}{\pi} - \left[\frac{x}{\pi}\right]\right)}}\,dx = \frac{\alpha\pi^3}{1+4\pi^2}$, $\alpha \in R$, where [x] is the greatest integer less than or equal to x, then the value of α is : **[JEE(Main)July2021]**

यदि $\int_0^{100\pi} \frac{\sin^2 x}{e^{\left(\frac{x}{\pi} - \left[\frac{x}{\pi}\right]\right)}}\,dx = \frac{\alpha\pi^3}{1+4\pi^2}$, $\alpha \in R$ है, जबकि [x], महत्तम पूर्णांक $\le x$ है, तो α बराबर है :

[JEE(Main)July2021]

(A) $200\,(1 - e^{-1})$ (B) $150\,(e^{-1} - 1)$ (C) $50\,(e - 1)$ (D) $100\,(1 - e)$

Ans. A

Sol. $$\int_0^{100\pi} \frac{\sin^2 x}{e^{\left\{\frac{x}{\pi}\right\}}}\,dx$$

Let $f(x) = \frac{\sin^2 x}{e^{\left\{\frac{x}{\pi}\right\}}}$

Period of f(x) = LCM $(\pi, \pi) = \pi$

$$\int_0^{100\pi} \frac{\sin^2 x}{e^{\left\{\frac{x}{\pi}\right\}}}\,dx = 100\int_0^{\pi} \frac{\sin^2 x}{e^{\left\{\frac{x}{\pi}\right\}}}\,dx$$

$$= 100\int_0^{\pi} \frac{\sin^2 x\,dx}{e^{\left\{\frac{x}{\pi}\right\}}}$$

$x = \pi t$

$dx = \pi dt$

$$= 100\pi\int_0^1 \frac{\sin^2 \pi t}{e^t}$$

$$100\pi\int_0^1 e^{-t}\left(\frac{1-\cos 2\pi t}{2}\right)dt$$

$$=50\pi\left[\int_0^1 e^{-t}dt-\int_0^1 e^{-t}\cos 2\pi t dt\right]$$

$$=50\pi\left[\left(1-\frac{1}{e}\right)-\left(\frac{e^{-t}}{4\pi^2+1}\left(-1\cos 2\pi t+2\pi\sin 2\pi t\ \right)\right)\right]$$

$$=50\pi\left[\left(1-\frac{1}{e}\right)-\left[\frac{-e^{-1}}{1+4\pi^2}+\frac{1}{1+4\pi^2}\right]\right]$$

$$=50\pi\left[\left(1-\frac{1}{e}\right)-\frac{1}{1+4\pi^2}\left(1-\frac{1}{e}\right)\right]$$

$$=50\pi\left(1-\frac{1}{e}\right)\left(\frac{4\pi^2}{1+4\pi^2}\right)$$

$$=\frac{200\left(1-\frac{1}{e}\right)\pi^3}{1+4\pi^2}=\frac{\alpha\pi^3}{1+4\pi^2}$$

By Comparision

$\alpha = 200\ (1-e^{-1})$

8. The value of the definite integral $\int_{\pi/24}^{5\pi/24}\frac{dx}{1+\sqrt[3]{\tan 2x}}$ is : **[JEE(Main)July2021]**

निश्चित समाकलन $\int_{\pi/24}^{5\pi/24}\frac{dx}{1+\sqrt[3]{\tan 2x}}$ बराबर है : **[JEE(Main)July2021]**

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{18}$ (C) $\frac{\pi}{12}$ (D) $\frac{\pi}{3}$

Ans. C

Sol. $I=\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}}\frac{dx}{1+(\tan 2x)^{1/3}}$ .....(1)

Apply king property $I=\int_a^b I(a+b-x)$

$$I = \int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1+(\cot 2x)^{1/3}} \qquad \text{.....(2)}$$

equation (1) + (2)

$$2I = \int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} dx = \left(\frac{5\pi}{24} - \frac{\pi}{24}\right) = \frac{4\pi}{24} = \frac{\pi}{6}$$

$$I = \frac{\pi}{12}$$

9. Let $f: [0, \infty) \to [0, \infty)$

be defined as $f(x) = \int_0^x [y]\,dy$ where [x] is the greatest integer less than or equal to x. Which of the following is true?

[JEE(Main)July2021]

(A) f is differentiable at every point in $[0, \infty)$.

(B) f is continuous everywhere except at the integer points in $[0, \infty)$.

(C) f is continuous at every point in $[0, \infty)$ and differentiable except at the integer points.

(D) f is both continuous and differentiable except at the integer points in $[0, \infty)$.

माना $f: [0, \infty) \to [0, \infty)$

$f(x) = \int_0^x [y]\,dy$

द्वारा परिभाषित है, जबकि [x] महत्तम पूर्णांक $\le x$ है। तो निम्न में से कौन–सा कथन सत्य है ? **[JEE(Main)July2021]**

(A) $[0, \infty)$ के प्रत्येक बिन्दु पर f अवकलनीय है

(B) पूर्णांक बिदुओं के अतिरिक्त $[0, \infty)$ के सभी बिन्दुओं पर f संतत है

(C) $[0, \infty)$ प्रत्येक बिन्दु पर f संतत और पूर्णांक बिन्दुओं के अतिरिक्त सभी बिन्दुओं पर अवकलनीय है

(D) पूर्णांक बिन्दुओं के अतिरिक्त $[0, \infty)$ के सभी बिन्दुओं पर f संतत तथा अवकलनीय दोनों है

Ans. C

Sol. Let $n \le x\ n + 1$

$$f(x) = \int_0^1 (0)\,dy + \int_1^1 (1)\,dy + \int_2^3 2dy ..... + \int_n^x (n)\,dy$$

$f(x) = 0 + 1 + 2\,(1) + 3\,(1)$ .....$(n-1)\,(1) + n\,(x-n)$

$= 1 + 2 + 3$ .....$+ n - 1 + nx - n^2$

$$= \frac{(n-1)(n)}{2} + nx - n^2$$

$$= n\left(x - \frac{n+1}{2}\right)$$

$$f(x)=\begin{cases} x\left(\dfrac{x-1}{2}\right) & x \in n \text{ (integer)} \\ [x]\left(\dfrac{2x-[x]-1}{2}\right) & x \text{ (integer)} \end{cases}$$

$\lim\limits_{x\to n} f(x)=\dfrac{n(n-1)}{2}$ continuous

$f(x)=\int\limits_0^x [y]\,dy$

$f'(x)=[x]$

ND at all integers

10. If $f(x)=\begin{cases}\int\limits_0^x (5+|1-t|)\,dt, & x>2 \\ 5x+1 & x\le 2\end{cases}$, then : **[JEE(Main)July2021]**

(A) f(x) is not differentiable at x = 1

(B) f(x) is not continuous at x = 2

(C) f(x) is everywhere differentiable

(D) f(x) is continuous but not differentiable at x = 2

यदि $f(x)=\begin{cases}\int\limits_0^x (5+|1-t|)\,dt, & x>2 \\ 5x+1 & x\le 2\end{cases}$ है, तो : **[JEE(Main)July2021]**

(A) x = 1 पर f(x) अवकलनीय नहीं है

(B) x = 2 पर f(x) संतत नहीं है

(C) f(x) प्रत्येक x के लिए अवकलनीय है

(D) x = 2 पर f(x) संतत है परन्तु अवकलनीय नहीं है

Ans. D

Sol. $f(x)=\begin{cases}\int\limits_0^x (5+|1-t|)\,dt & x>2 \\ 5x+1 & x\le 2\end{cases}$

$f(x)=\int_0^1 (5+1-t)\,dt+\int_1^x (5+t-1)\,dt$

$=\int_0^1 (6-t)\,dt+\int_1^x (t+4)\,dt$

$=6-\dfrac{1}{2}+\dfrac{x^2}{2}+4x-\dfrac{1}{2}-4$

$$f(x)=\frac{11+x^2+8x-1-8}{2}=\frac{x^2+8x+2}{2}$$

$$f(x)=\begin{cases}\frac{x^2+8x+2}{2} & x>2\\ 5x+1 & x\le 2\end{cases}$$

$$f(2^+)=\frac{4+16+2}{2}=11$$

$f(2^-)=11$

$f(2)=11$

Continous

$$f'(x)=\begin{cases}\frac{2x+8}{2} & x>2\\ 5 & x<2\end{cases}$$

$$f'(x)=\begin{cases}x+4 & x>2\\ 5 & x<2\end{cases}$$

$f'(2^+)=6 \qquad f'(2^-)=2$

Not differentiable

11. The value of the integral $\int_{-1}^{1}\log\left(x+\sqrt{x^2+1}\right)dx$ is : **[JEE(Main)July2021]**

समाकलन $\int_{-1}^{1}\log\left(x+\sqrt{x^2+1}\right)dx$ का मान है : **[JEE(Main)July2021]**

(A) 0 (B) 2 (C) -1 (D) 1

Ans. A

Sol. $$I=\int_{-1}^{1}\log\left(x+\sqrt{x^2+1}\right)dx$$

$$I=\int_{0}^{1}\left(\log(x+\sqrt{x^2+1})+\log(-x+\sqrt{x^2+1})\right)dx$$

$$=\int_0^1\log\left(x^2+1-x^2\right)dx=0$$

12. The value of is equal to $\lim_{n\to\infty}\frac{1}{n}\sum_{j=1}^{n}\frac{(2j-1)+8n}{(2j-1)+4n}$ is equal to : **[JEE(Main)July2021]**

$\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{j=1}^{n}\frac{(2j-1)+8n}{(2j-1)+4n}$ का मान बराबर है : **[JEE(Main)July2021]**

(A) $5+\log_e\left(\frac{3}{2}\right)$ (B) $1+2\log_e\left(\frac{3}{2}\right)$ (C) $3+2\log_e\left(\frac{2}{3}\right)$ (D) $2-\log_e\left(\frac{2}{3}\right)$

Ans. B

Sol. $\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{j=1}^{n}\frac{2\frac{j}{n}-\frac{1}{n}+8}{2\frac{j}{n}-\frac{1}{n}+4}=\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{j=1}^{n}\frac{2\frac{j}{n}+8}{2\frac{j}{n}+4}$ $\left(\lim\limits_{n\to\infty}\frac{1}{n}=0\right)$

$$=\int_0^1\frac{2x+8}{2x+4}dx=\int_0^1\left(1+\frac{4}{2x+4}\right)dx=x+\ell n(x+2)\Big|_0^1$$

$$=1+2\,ln\left(\frac{3}{2}\right)$$

13. The value of definite integral $\int\limits_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{dx}{\left(1+e^{x\cos x}\right)\left(\sin^4 x+\cos^4 x\right)}$ is equal to : **[JEE(Main)July2021]**

निश्चित समाकलन $\int\limits_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{dx}{\left(1+e^{x\cos x}\right)\left(\sin^4 x+\cos^4 x\right)}$ का मान बराबर है : **[JEE(Main)July2021]**

(A) $-\frac{\pi}{4}$ (B) $\frac{\pi}{\sqrt{2}}$ (C) $-\frac{\pi}{2}$ (D) $\frac{\pi}{2\sqrt{2}}$

Ans. D

Sol. $I=\int\limits_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{dx}{\left(1+e^{x\cos x}\right)\left(\sin^4 x+\cos^4 x\right)}$ .....(1)

Using $\int\limits_a^b f(x)\,dx=\int\limits_a^b f(a+b-x)\,dx$

$I=\int\limits_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{dx}{\left(1+e^{x\cos x}\right)\left(\sin^4 x+\cos^4 x\right)}$ .....(2)

$$2I = \int_{\frac{-\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{\sin^4 x + \cos^4 x} = 2 \int_0^{\frac{\pi}{4}} \frac{dx}{\sin^4 x + \cos^4 x}$$

$$I = \int_0^{\frac{\pi}{4}} \frac{\left(1+\tan^2 x\right)\sec^2 x}{\left(1+\tan^4 x\right)} dx$$

$\tan x = t$

$\sec^2 x\, dx = dt$

$$I = \int_0^1 \frac{1+t^2}{1+t^4} dt = \int_0^1 \frac{\left(1+\frac{1}{t^2}\right)}{t^2+\frac{1}{t^2}} dt$$

$$I = \int_0^1 \frac{\left(1+\frac{1}{t^2}\right)dt}{\left(t-\frac{1}{t}\right)^2+2}$$

$$I = \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{t-\frac{1}{t}}{\sqrt{2}}\right)\Bigg|_0^1 = \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{t^2-1}{\sqrt{2}t}\right)\Bigg|_0^1 = \frac{\pi}{2\sqrt{2}}$$

14. Let $f: (a, b) \to R$ be twice differentiable function such that $f(x) = \int_a^x g(t)\,dt$ for a differentiable function g(x).

If $f(x) = 0$ has exactly five distinct roots in (a, b), then g(x)g'(x)=0 has at least : **[JEE(Main)July2021]**

(A) seven roots in (a, b) (B) twelve roots in (a, b)

(C) five roots in (a, b) (D) three roots in (a, b)

माना फलन $f: (a, b) \to R$ दो बार अवकलनीय है, जबकि एक अवकलनीय फलन g(x) के लिए $f(x) = \int_a^x g(t)\,dt$ है। यदि (a, b) में $f(x) = 0$ के ठीक पाँच भिन्न मूल हैं, तो g(x)g'(x)=0 के (a, b) में कम से कम : **[JEE(Main)July2021]**

(A) (a, b) में सात मूल हैं (B) (a, b) में बारह मूल हैं

(C) (a, b) में पाँच मूल हैं (D) (a, b) में तीन मूल हैं

Ans. A

Sol. $f'(x) = g(x)$

$f''(x) = g'(x)$

$g(x) \cdot g'(x) = 0$

$f'(x) \cdot f''(x) = 0$ .....(1)

Given f(x) = 0 has 5 roots in (a, b)

$\therefore$ If f(x) = 0 has at least 5 roots

$\Rightarrow$ f'(x) = 0 has atleast 4 roots

$\Rightarrow$ f''(x) = 0 has atleast 3 roots

f'(x) . f''(x) = 0 has atleast 7 roots

g(x). g'(x) = 0 has atleast 7 roots.

15. Let the domain of the function $f(x) = \log_4\left(\log_5\left(\log_3\left(18x - x^2 - 77\right)\right)\right)$ be (a, b). Then the value of the integral $\int_a^b \frac{\sin^3 x}{\left(\sin^3 x + \sin^3\left(a + b - x\right)\right)} dx$ is equal to. **[JEE(Main)July2021]**

माना फलन $f(x) = \log_4\left(\log_5\left(\log_3\left(18x - x^2 - 77\right)\right)\right)$ का प्रांत (a, b) है। समाकलन $\int_a^b \frac{\sin^3 x}{\left(\sin^3 x + \sin^3\left(a + b - x\right)\right)} dx$ का मान बराबर है ———— । **[JEE(Main)July2021]**

Ans. 1

Sol. $\log_5(\log_3(18x - x^2 - 77) > 0$

$\Rightarrow \log_3(18x - x^2 - 77) > 1$

$\Rightarrow 18x - x^2 - 77 > 3$

$\Rightarrow x^2 - 18x + 80 < 0$

and $\log_3\left(18x - x^2 - 77\right) > 0$

$18x - x^2 - 77 > 1$

$x^2 - 18x + 77 < -1$

$x^2 - 18x + 78 < 0$

$x \in \left(9 - \sqrt{3}, 9 + \sqrt{3}\right)$

$x \in (8, 10)$

and $18x - x^2 - 77 > 0$

$\Rightarrow x \in (7, 11)$

$$I = \int_8^{10} \frac{\sin^3 x}{\sin^3 x + \sin^3(18 - x)} dx \quad \text{..... (i)}$$

Use $\int_a^b f(x)\,dx = \int_a^b f\left(a + b - x\right)dx$

$$I = \int_8^{10} \frac{\sin^3 x(18 - x)}{\sin^3(18 - x) + \sin^3 x} dx \quad \text{..... (ii)}$$

(i) + (ii)

$2I = \int_8^{10} dx = 2$

$I = 1$

16. Let F : [3, 5] $\to$ R be a twice differentiable function on (3,5) such that $F(x) = e^{-x}\int_3^x \left(3t^2 + 2t + 4F'(t)\right)dt$. If $F'(4) = \frac{\alpha e^{\beta} - 224}{\left(e^{\beta} - 4\right)^2}$, then $\alpha + \beta$ is equal to_____. **[JEE(Main)July2021]**

माना F : [3, 5] $\to$ R, (3,5) पर दो बार अवकलनीय फलन है, जिसके लिए $F(x) = e^{-x}\int_3^x \left(3t^2 + 2t + 4F'(t)\right)dt$ है। यदि $F'(4) = \frac{\alpha e^{\beta} - 224}{\left(e^{\beta} - 4\right)^2}$ है, तो $\alpha + \beta$ बराबर है_____ **[JEE(Main)July2021]**

Ans. 16

Sol. $e^x F(x) = \int_3^x \left(3t^2 + 2t + 4F'(t)\right)dt$ .....(1)

$e^x F'(x) + e^x F(x) = 3x^2 + 2x + 4F'(x)$

Put $x = 4$

$e^4 F'(4) + e^4 F(4) = 48 + 8 + 4F'(4)$

$F'(4) = \frac{56 - e^4 F(4)}{e^4 - 4}$

Now put $x = 4$ in equation(1)

$e^4 F(4) = \int_3^4 \left(3t^2 + 2t + 4F'(t)\right)dt$

$e^4 F(4) = t^3 + t^2 + 4F(t)\Big|_3^4$

$e^4 F(4) = (4^3 - 3^3) + (4^2 - 3^2) + 4[F(4) - F(3)]$ $\{\because F(3) = 0\}$

$(e^4 - 4) F(4) = 44$

$F'(4) = \frac{56 - e^4\left(\frac{44}{e^4 - 4}\right)}{\left(e^4 - 4\right)} = \frac{12e^4 - 224}{\left(e^4 - 4\right)^2}$

$\alpha = 12, \beta = 4 \Rightarrow \alpha + \beta = 16$

17. If $\int_0^{\pi} (\sin^3 x)\, e^{-\sin^2 x}\, dx = \alpha - \frac{\beta}{e}\int_0^1 \sqrt{t}\, e^t dt$, then $\alpha + \beta$ is equal to _____. **[JEE(Main)July 2021]**

यदि $\int_0^{\pi} (\sin^3 x)\, e^{-\sin^2 x}\, dx = \alpha - \frac{\beta}{e}\int_0^1 \sqrt{t}\, e^t dt$ है, तो $\alpha + \beta$ बराबर है _____.

[JEE(Main)July2021]

Ans. 5

Sol. $\int_0^{\pi} (\sin^3 x)\, e^{-\sin^2 x}\, dx = \alpha - \frac{\beta}{e}\int_0^1 \sqrt{t}\, e^t dt,$

$$I = \int_0^{\pi} \sin^3 x\, e^{-\sin^2 x} dx$$

$$I = 2\int_0^{\pi/2} \sin^3 x\, e^{-\sin^2 x} dx$$

$$I = 2\left[\int_0^{\pi/2} \left(1-\cos^2 x\right)\sin x\, e^{\cos^2 x - 1} dx\right]$$

Substitute $\cos^2 x = t$

$\Rightarrow -2\cos x \sin x\, dx = dt$

$$\Rightarrow \sin x\, dx = \frac{-dt}{2\cos x} = \frac{-dt}{2\sqrt{t}}$$

$$I = 2\left[\int_1^0 (1-t)e^{t-1}\left(\frac{-dt}{2\sqrt{t}}\right)\right]$$

$$= \int_0^1 \frac{(1-t)}{\sqrt{t}} e^{t-1}\, dt$$

$$= \frac{1}{e}\int_0^1 \frac{(1-t)}{\sqrt{t}} e^t\, dt$$

$$= \frac{1}{e}\left[\int_0^1 \frac{1}{\sqrt{t}} e^t dt - \int_0^1 \sqrt{t}\, e^t dt\right]$$

$$= \frac{1}{e}\left[\left[2\sqrt{t}\, e^t\right]_0^1 - \int_0^1 2\sqrt{t}\, e^t\, dt - \int_0^1 \sqrt{t}\, e^t\, dt\right]$$

$$= \frac{1}{e}\left[2e - 3\int_0^1 \sqrt{t}\, e^t dt\right] = 2 - \frac{3}{e}\int_0^1 \sqrt{t}\, e^t dt$$

$\alpha = 2,\ \beta = 3$

$\Rightarrow \alpha + \beta = 5$

18. The integral $\int \frac{1}{\sqrt[4]{(x-1)^3(x+2)^5}}\,dx$ is equal to : (where C is a constant of integration)

[JEE(Main)Aug.2021]

समाकलन $\int \frac{1}{\sqrt[4]{(x-1)^3(x+2)^5}}\,dx$ का बराबर है : (जहाँ C एक समाकलन अचर है) **[JEE(Main)Aug.2021]**

(A) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{\frac{1}{4}}+C$ (B) $\frac{3}{4}\left(\frac{x+2}{x-1}\right)^{\frac{1}{4}}+C$

(C) $\frac{3}{4}\left(\frac{x-1}{x+2}\right)^{\frac{5}{4}}+C$ (D) $\frac{3}{4}\left(\frac{x+2}{x-1}\right)^{\frac{5}{4}}+C$

Ans. A

Sol. $$I=\int \frac{dx}{\sqrt[4]{(x-1)^8.\frac{(x+2)^5}{(x-1)^5}}}$$

$$I=\int \frac{dx}{(x-1)^2\left(\frac{x+2}{x-1}\right)^{\frac{5}{4}}}$$

Let, $\frac{x+2}{x-1}=t$

$$\Rightarrow \frac{-3\,dx}{(x-1)^2}=dt$$

$$I=\frac{-1}{3}\int \frac{dt}{t^{5/4}}=\frac{-1}{3}.\frac{t^{-1/4}}{\frac{-1}{4}}+C$$

$$I=\frac{4}{3}\left(\frac{x+2}{x-1}\right)^{-1/4}+C$$

$$I=\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4}+C$$

19. If $\int \frac{dx}{(x^2+x+1)^2} = a\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + b\left(\frac{2x+1}{x^2+x+1}\right) + C, x > 0$ where C is the constant of integration, then the value of $9(\sqrt{3}a+b)$ is equal to_____. **[JEE(Main)Aug.2021]**

यदि $\int \frac{dx}{(x^2+x+1)^2} = a\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + b\left(\frac{2x+1}{x^2+x+1}\right) + C, x > 0$ है, जहाँ C समाकलन अचर है, तो $9(\sqrt{3}a+b)$ का मान बराबर है _____ **[JEE(Main)Aug.2021]**

Ans. 15

Sol. $$I = \int \frac{dx}{\left(\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2\right)^2}$$

Let $x + \frac{1}{2} = \frac{\sqrt{3}}{2}\tan\theta$

$$dx = \frac{\sqrt{3}}{2}\sec^2\theta\, d\theta$$

$$I = \frac{\sqrt{3}}{2}\int \frac{\sec^2\theta d\theta}{\frac{9}{16}\sec^4\theta}$$

$$I = \frac{8}{3\sqrt{3}}\int\left(\frac{1+\cos 2\theta}{2}\right)d\theta$$

$$I = \frac{4}{3\sqrt{3}}\left[\theta + \frac{\sin 2\theta}{2}\right] + C$$

$$I = \frac{4}{3\sqrt{3}}\left[\theta + \frac{\tan\theta}{1+\tan^2\theta}\right] + C$$

$$I = \frac{4}{3\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + \frac{1}{3}\left(\frac{2x+1}{x^2+x+1}\right) + C$$

$$a = \frac{4}{3\sqrt{3}}$$

$$b = \frac{1}{3}$$

$$9(\sqrt{3}a+b) = 15$$

20. If $\int \frac{2e^x + 3e^{-x}}{4e^x + 7e^{-x}} dx = \frac{1}{14}(ux + v\log_e(4e^x + 7e^{-x})) + C$, where C is as constant of integration, then u + v is equal to _____. **[JEE(Main)Aug.2021]**

यदि $\int \frac{2e^x + 3e^{-x}}{4e^x + 7e^{-x}} dx = \frac{1}{14}(ux + v\log_e(4e^x + 7e^{-x})) + C$ है, जहाँ C एक समाकलन अचर है, तो u + v बराबर है_____ **[JEE(Main)Aug.2021]**

Ans. 7

Sol. $I = \int \frac{2e^x + 3e^{-x}}{4e^x + 7e^{-x}} dx$

$2e^x + 3e^{-x} = A(4e^x + 7e^{-x}) + B(4e^x - 7e^{-x})$

By comparing the cofficients on both hids

$A + B = \frac{1}{2}$

$A - B = \frac{3}{7}$

$A = \frac{13}{28}$

$B = \frac{1}{28}$

$I = \int \frac{\frac{13}{28}(4e^x + 7e^{-x}) + \frac{1}{28}(4e^x - 7e^{-x})}{4e^x + 7e^{-x}}$

$I = \frac{13}{28}x + \frac{1}{28}\ln(4e^x + 7e^{-x}) + C$

$I = \frac{1}{14}\left(\frac{13}{2}x + \frac{1}{2}\ln(4e^x + 7e^{-x})\right) + C$

$u = \frac{13}{2}$

$v = \frac{1}{2}$

$u + v = 7$

21. If $\int \frac{\sin x}{\sin^3 x + \cos^3 x} dx = \alpha \log_e |1 + \tan x| + \beta \log_e |1 - \tan x + \tan^2 x| + \gamma \tan^{-1}\left(\frac{2\tan x - 1}{\sqrt{3}}\right) + C$, when C is constant of integration, then the value of $18(\alpha + \beta + \gamma^2)$ is_____. **[JEE(Main)Aug.2021]**

यदि $\int \frac{\sin x}{\sin^3 x+\cos^3 x}dx = \alpha \log_e|1+\tan x| + \beta \log_e|1-\tan x+\tan^2 x| + \gamma \tan^{-1}\left(\frac{2\tan x-1}{\sqrt{3}}\right)+C$, जहाँ C एक समाकलन अचर है, तो $18(\alpha+\beta+\gamma^2)$ का मान बराबर है_____ **[JEE(Main)Aug.2021]**

Ans. 3

Sol. $\int \frac{\sin x}{\sin^3 x+\cos^3 x}dx$

$\int \frac{\tan x \sec^2 x}{\tan^3 x+1}dx$

$\tan x = t$

$\sec^2 x dx = dt$

$\int \frac{tdt}{t^3+1}$

$\int \frac{tdt}{(t+1)(t^2-t+1)}$

$\frac{t}{(t+1)(t^2-t+1)} = \frac{A}{t+1}+\frac{Bt+c}{t^2-t+1}$

$t = A(t^2-t+1)+(t+1)(Bt+c)$

$t = At^2-At+A+Bt^2+Ct+Bt+c$

$t = (A+B)t^2+(B+C-A)t+A+C$

$A+B=0$ $\quad B+C-A=1$ $\quad A+C=0$

$B+C-A=1$ $\quad C=-A$

$-A-A-A=1$ $\quad B=-A$

$A=-\frac{1}{3}$ $\quad C=\frac{1}{3}$ $\quad B=\frac{1}{3}$

$\int \frac{tdt}{(t+1)(t^2-t+1)} = -\frac{1}{3}\int \frac{dt}{t+1}+\frac{1}{3}\int \frac{t+1}{t^2-t+1}dt = -\frac{1}{3}\ln|t+1|+\frac{1}{3}\int \frac{t+1}{t^2-t+1}dt$

$\int \frac{t+1}{t^2-t+1}dt = \frac{1}{2}\int \frac{2t}{t^2-t+1}dt+\int \frac{dt}{t^2-t+1}$

$= \frac{1}{2}\int \frac{2t-1}{t^2-t+1}+\frac{3}{2}\int \frac{dt}{t^2-t+1}$

$= \frac{1}{2}\ln|t^2-t+1|+\frac{3}{2}\int \frac{dt}{\left(t-\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$

$$= \frac{1}{2}\ln|t^2 - t + 1| + \frac{3}{2} \times \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{t - \frac{1}{\sqrt{2}}}{\frac{\sqrt{3}}{2}}\right)$$

$$= -\frac{1}{3}\ln|t + 1| + \frac{1}{6}\ln|t^2 - t + 1| + \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2t - 1}{\sqrt{3}}\right)$$

$$= -\frac{1}{3}\ln|1 + \tan x| + \frac{1}{6}\ln|\tan^2 x - \tan x + 1| + \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2\tan x - 1}{\sqrt{3}}\right)$$

$$\alpha = -\frac{1}{3} \qquad \beta = \frac{1}{6} \qquad \gamma = \frac{1}{\sqrt{3}}$$

$$18(\alpha + \beta + \gamma^2) = 18\left(-\frac{1}{3} + \frac{1}{6} + \frac{1}{3}\right)$$

$$= 18 \times \frac{1}{6} = 3$$

22. The value of $\lim\limits_{n\to\infty} \frac{1}{n} \sum\limits_{r=0}^{2n-1} \frac{n^2}{n^2 + 4r^2}$ is : **[JEE(Main)Aug.2021]**

$\lim\limits_{n\to\infty} \frac{1}{n} \sum\limits_{r=0}^{2n-1} \frac{n^2}{n^2 + 4r^2}$ का मान है : **[JEE(Main)Aug.2021]**

(A) $\frac{1}{4}\tan^{-1}(4)$ (B) $\frac{1}{2}\tan^{-1}(2)$ (C) $\frac{1}{2}\tan^{-1}(4)$ (D) $\tan^{-1}(4)$

Ans. C

Sol. $$\lim_{n\to\infty} \frac{1}{n} \sum_{r=0}^{2n-1} \frac{n^2}{n^2 + 4r^2} = \int_0^2 \frac{1}{1 + 4x^2}\,dx$$

$$= \frac{1}{2}\left(\tan^{-1}(2x)\right)_0^2$$

$$= \frac{\tan^{-1}(4)}{2}$$

23. The value of $\int\limits_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left(\left(\frac{x+1}{x-1}\right)^2 + \left(\frac{x-1}{x+1}\right)^2 - 2\right)^{\frac{1}{2}} dx$ is : **[JEE(Main)Aug.2021]**

$\int_{\frac{-1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}}\left(\left(\frac{x+1}{x-1}\right)^2+\left(\frac{x-1}{x+1}\right)^2-2\right)^{\frac{1}{2}}dx$ का मान है : **[JEE(Main)Aug.2021]**

(A) $4\log_e(3+2\sqrt{2})$ (B) $2\log_e 16$ (C) $\log_e 4$ (D) $\log_e 16$

Ans. D

Sol. $I=\int_{\frac{-1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}}f(x)dx$

$f(x)=\left(\left(\frac{x+1}{x-1}\right)^2+\left(\frac{x-1}{x+1}\right)^2-2\right)^{\frac{1}{2}}$.

$=\sqrt{\left(\frac{x+1}{x-1}-\frac{x-1}{x+1}\right)^2}$

$f(x)=\left|\frac{x+1}{x-1}-\frac{x-1}{x+1}\right|$

$f(x)=f(-x)$

$I=\int_{\frac{-1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}}f(x)dx$

$=2\int_0^{\frac{1}{\sqrt{2}}}f(x)dx$

$=2\int_0^{\frac{1}{\sqrt{2}}}\left|\frac{x+1}{x-1}-\frac{x-1}{x+1}\right|dx$

$=2\int_0^{\frac{1}{\sqrt{2}}}\left(\frac{x-1}{1+x}-\frac{x+1}{x-1}\right)dx$

$=2\int_0^{\frac{1}{\sqrt{2}}}\left(1-\frac{2}{1+x}\right)-\left(1+\frac{2}{x-1}\right)dx$

$=2\left[-2\ln|1+x|-2\ln|1-x|\right]_0^{\frac{1}{\sqrt{2}}}$

$= -4\ln\left(1+\frac{1}{\sqrt{2}}\right) - 4\ln\left(1-\frac{1}{\sqrt{2}}\right)$

$= -4\ln\left(1-\frac{1}{2}\right) = -4\ln\left(\frac{1}{2}\right) = 4\ln 2$

$= \ln 16$

24. If the value of the integeral $\int_0^5 \frac{x+[x]}{e^{x-[x]}}\,dx = \alpha e^{-1} + \beta$, where $\alpha, \beta, \in R$, $5\alpha + 6\beta = 0$, and [x] denotes the greatest integer less than or equal to x; then the value of $(\alpha+\beta)^2$ is equal to : **[JEE(Main)Aug.2021]**

यदि समाकलन $\int_0^5 \frac{x+[x]}{e^{x-[x]}}\,dx = \alpha e^{-1} + \beta$ है, जहाँ $\alpha, \beta, \in R$, $5\alpha + 6\beta = 0$ तथा [x] महत्तम पूर्णांक हैं, तो $\leq x$ हैं, तो $(\alpha+\beta)^2$ का मान बराबर है : **[JEE(Main)Aug.2021]**

(A) 100 (B) 16 (C) 36 (D) 25

Ans. D

Sol. $\int_0^1 \frac{x}{e^x}dx + \int_1^2 \frac{x+1}{e^{x-1}}dx + \int_2^3 \frac{x+2}{e^{x-2}} + \int_3^4 \frac{x+3}{e^{x-3}}dx + \int_4^5 \frac{x+4}{e^{x-4}}dx$

$x = a+1 \quad x = b+2 \quad x = c+3 \quad x = v+4$

$= \int_0^1 \frac{x}{e^x}dx + \int_0^1 \frac{a+2}{e^a}da + \int_0^1 \frac{b+4}{e^b}db + \int_0^1 \frac{c+6}{e^c}dc + \int_0^1 \frac{v+8}{e^v}dv$

$= \int_0^1 \frac{5x+20}{e^x} = 5\int_0^1 \frac{x+4}{e^x}dx = -30e^{-1} + 25$

$\alpha = -30$

$\beta = 25$

$(\alpha+\beta)^2 = 25$

25. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1+\sin^2 x}{1+\pi^{\sin x}}\right)dx$ is : **[JEE(Main)Aug.2021]**

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1+\sin^2 x}{1+\pi^{\sin x}}\right)dx$ का मान है : **[JEE(Main)Aug.2021]**

(A) $\frac{3\pi}{4}$ (B) $\frac{3\pi}{2}$ (C) $\frac{\pi}{2}$ (D) $\frac{5\pi}{2}$

Ans. A

Sol. $I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{1+\sin^2 x}{1+\pi^{\sin x}}\right)$

Apply king property

$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{1+\sin^2 x}{1+\pi^{-\sin x}}\right)$

$\Rightarrow 2I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(1+\sin^2 x\right)dx$

$\Rightarrow I=\frac{3\pi}{4}$

26. If $U_n=\left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)^2\ldots\left(1+\frac{n^2}{n^2}\right)^n$, then $\lim_{n\to\infty}(U_n)^{\frac{-4}{n^2}}$ is equal to : **[JEE(Main)Aug.2021]**

यदि $U_n=\left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)^2\ldots\left(1+\frac{n^2}{n^2}\right)^n$ है, तो $\lim_{n\to\infty}(U_n)^{\frac{-4}{n^2}}$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $\frac{16}{e^2}$ (B) $\frac{4}{e}$ (C) $\frac{4}{e^2}$ (D) $\frac{e^2}{16}$

Ans. D

Sol. $y=\lim_{n\to\infty}\left\{\left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)^2----\left(1+\frac{n^2}{n^2}\right)^n\right\}^{\frac{-4}{n^2}}$

$y=\lim_{n\to\infty}\left(\prod_{r=1}^{n}\left(1+\frac{r^2}{n^2}\right)^r\right)^{\frac{-4}{n^2}}$

$\ln(y)=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{-4r}{n^2}\ln\left(1+\frac{r^2}{n^2}\right)$

$\ln(y)=\int_0^1-4x\ln\left(1+x^2\right)dx$

Let $1 + x^2 = t$

$2x\,dx = dt$

$$\ln y = \int_1^2 -2\ln t\,dt = -2[t\ln t - t]_1^2$$

$\ln y = 2 - \ln 16$

$y = e^{2-\ln 16}$

$$y = \frac{e^2}{16}$$

27. $\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e(x^2 - 44x + 484)}dx$ is equal to : **[JEE(Main)Aug.2021]**

$\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e(x^2 - 44x + 484)}dx$ के बराबर है : **[JEE(Main)Aug.2021]**

(A) 8 (B) 6 (C) 10 (D) 5

Ans. D

Sol. $$I = \int_6^{16} \frac{\log_e x^2 dx}{\log_e x^2 + \log_e(22 - x)^2} \ldots\ldots(i)$$

Apply king property

$$I = \int_6^{16} \frac{\log_e(22 - x)^2}{\log_e(22 - x)^2 + \log_e x^2} \ldots\ldots(ii)$$

$$2I = \int_6^{16} dx = 10$$

$I = 5$

28. The value of the integral $\int_0^1 \frac{\sqrt{x}dx}{(1+x)(1+3x)(3+x)}$ is : **[JEE(Main)Aug.2021]**

समाकलन $\int_0^1 \frac{\sqrt{x}dx}{(1+x)(1+3x)(3+x)}$ का मान है : **[JEE(Main)Aug.2021]**

(A) $\frac{\pi}{8}\left(1 - \frac{\sqrt{3}}{2}\right)$ (B) $\frac{\pi}{4}\left(1 - \frac{\sqrt{3}}{2}\right)$ (C) $\frac{\pi}{4}\left(1 - \frac{\sqrt{3}}{6}\right)$ (D) $\frac{\pi}{8}\left(1 - \frac{\sqrt{3}}{6}\right)$

Ans. A

Sol. $\int_0^1 \frac{\sqrt{x}dx}{(1+x)(1+3x)(3+x)}$ $x = t^2$

$I = \int_0^1 \frac{2t^2dt}{(1+t^2)(1+3t^2)(3+t^2)}$ $dx = 2tdt$

$I = \int_0^1 \frac{(3t^2+1)-(t^2+1)dt}{(t^2+1)(3t^2+1)(t^2+3)}$

$I = \int_0^1 \frac{dt}{(1+t^2)(t^2+3)} - \int_0^1 \frac{dt}{(3t^2+1)(t^2+3)}$

$I = \frac{1}{2}\int_0^1 \frac{(t^2+3)-(t^2+1)}{(1+t^2)(t^2+3)}dt + \frac{1}{8}\int_0^1 \frac{(1+3t^2)-3(3+t^2)}{(3t^2+1)(t^2+3)}dt$

$I = \frac{1}{2}\int_0^1 \frac{dt}{1+t^2} - \frac{1}{2}\int_0^1 \frac{dt}{t^2+3} + \frac{1}{8}\int_0^1 \frac{dt}{t^2+3} - \frac{3}{8}\int_0^1 \frac{dt}{3t^2+1}$

$I = \frac{1}{2}\int_0^1 \frac{dt}{1+t^2} - \frac{3}{8}\int_0^1 \frac{dt}{t^2+3} - \frac{3}{8}\int_0^1 \frac{dt}{1+3t^2}$

$I = \frac{1}{2}\tan^{-1}\Big]_0^1 - \frac{3}{8\sqrt{3}}\tan^{-1}\left(\frac{t}{\sqrt{3}}\right)\Big]_0^1 - \frac{3}{8\sqrt{3}}\tan^{-1}(\sqrt{3}t)\Big]_0^1$

$I = \frac{1}{2}\left(\frac{\pi}{4}-0\right) - \frac{3}{8\sqrt{3}}\left(\frac{\pi}{6}\right) - \frac{3}{8\sqrt{3}}\left(\frac{\pi}{3}\right)$

$I = \frac{\pi}{8} - \frac{\sqrt{3}}{16}\pi$

$I = \frac{\pi}{8}\left(1-\frac{\sqrt{3}}{2}\right)$

29. If [x] is the greatest integer $\le$ x, then $\pi^2\int_0^2 \left(\sin\frac{\pi x}{2}\right)\left(x-[x]\right)^{[x]}dx$ is equal to : **[JEE(Main)Aug.2021]**

यदि [x] महत्तम पूर्णांक $\le$ x है, तो $\pi^2\int_0^2 \left(\sin\frac{\pi x}{2}\right)\left(x-[x]\right)^{[x]}dx$ बराबर है : **[JEE(Main)Aug.2021]**

(A) 4 (π + 1) (B) 2 (π + 1) (C) 4 (π – 1) (D) 2 (π – 1)

Ans. C

Sol. $\pi^2\int_0^2\left(\sin\frac{\pi x}{2}\right)(x-[x])^{[x]}dx$

$=\pi^2\left[\int_0^1\sin\frac{\pi x}{2}dx+\int_1^2(x-1)\sin\frac{\pi x}{2}dx\right]$

$=\pi^2\left[-\frac{2}{\pi}\cos\frac{\pi x}{2}\Big|_0^1\left((x-1)\left(-\frac{2}{\pi}\cos\frac{\pi x}{2}\right)+\int\frac{2}{\pi}\cos\frac{\pi x}{2}dx\right)\right]$

$=\pi^2\left[-\frac{2}{\pi}(0-1)-(x-1)\frac{2}{\pi}\cos\frac{\pi x}{2}\Big|_1^2+\frac{4}{\pi^2}\sin\frac{\pi x}{2}\Big|_1^2\right]$

$=\pi^2\left[\frac{2}{\pi}+\frac{2}{\pi}-\frac{4}{\pi^2}\right]$

$=\pi^2\left[\frac{4}{\pi}-\frac{4}{\pi^2}\right]=4\pi-4=4(\pi-1)$

30. Let $f:\mathbf{R}\to\mathbf{R}$ be a continuous function. Then $\lim\limits_{x\to\frac{\pi}{4}}\dfrac{\frac{\pi}{4}\int_2^{\sec^2 x}f(x)dx}{x^2-\frac{\pi^2}{16}}$ is equal to : **[JEE(Main)Aug.2021]**

माना $f:\mathbf{R}\to\mathbf{R}$ एक संतत फलन है। तब $\lim\limits_{x\to\frac{\pi}{4}}\dfrac{\frac{\pi}{4}\int_2^{\sec^2 x}f(x)dx}{x^2-\frac{\pi^2}{16}}$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $2f\left(\sqrt{2}\right)$ (B) $4f(2)$ (C) $2f(2)$ (D) $f(2)$

Ans. C

Sol. $\lim\limits_{x\to\frac{\pi}{4}}\dfrac{\frac{\pi}{4}\int_2^{\sec^2 x}f(x)dx}{x^2-\frac{\pi^2}{16}}\left(\frac{0}{0}\right)$

L' Hospital Rule

$=\lim\limits_{x\to\frac{\pi}{4}}\dfrac{\frac{\pi}{4}f(\sec^2 x).2\sec^2 x.\tan x}{2x}$

$$= \frac{\frac{\pi}{4}.f(2).2\times 2\times 1}{2\times\frac{\pi}{4}}$$

$$= 2f(2)$$

31. Let $J_{n,m} = \int_0^{\frac{1}{2}} \frac{x^n}{x^m - 1}dx$, $\forall$ $n > m$ and $n, m \in N$. Consider a matrix $A = [a_{ij}]_{3\times 3}$ where $a_{ij} = \begin{cases} J_{6+i,3} - J_{i+3,3}, & i \le j \\ 0, & i > j \end{cases}$.

Then $|\text{adj}\, A^{-1}|$ is : **[JEE(Main)Aug.2021]**

माना सभी $n, m \in N$, $n > m$ के लिए $J_{n,m} = \int_0^{\frac{1}{2}} \frac{x^n}{x^m - 1}dx$ है। एक आव्यूह $A = [a_{ij}]_{3\times 3}$ जहाँ $a_{ij} = \begin{cases} J_{6+i,3} - J_{i+3,3}, & i \le j \\ 0, & i > j \end{cases}$ है, का

विचार कीजिए। तब $|\text{adj}\, A^{-1}|$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $(105)^2 \times 2^{38}$ (B) $(15)^2 \times 2^{42}$ (C) $(15)^2 \times 2^{34}$ (D) $(105)^2 \times 2^{36}$

Ans. A

Sol. $J_{n,m} = \int_0^{\frac{1}{2}} \frac{x^n}{x^m - 1}dx, \quad \forall n > m$

$a_{ij} = J_{6+i,3} - J_{i+3,3}$

$$= \int_0^{\frac{1}{2}} \frac{x^{6+i}}{x^3 - 1}dx - \int_0^{\frac{1}{2}} \frac{x^{3+i}}{x^3 - 1}dx$$

$$= \int_0^{\frac{1}{2}} \frac{x^{3+i}(x^3 - 1)}{(x^3 - 1)}dx$$

$$= \int_0^{\frac{1}{2}} x^{3+i}dx = \left[\frac{x^{4+i}}{4+i}\right]_0^{\frac{1}{2}}$$

$$= \frac{1}{4+i}.\left(\frac{1}{2}\right)^{4+i} = \frac{1}{(4+i)2^{4+i}}$$

$$A = \begin{bmatrix} \frac{1}{5.2^5} & \frac{1}{5.2^5} & \frac{1}{5.2^5} \\ 0 & \frac{1}{6.2^6} & \frac{1}{6.2^6} \\ 0 & 0 & \frac{1}{7.2^7} \end{bmatrix}$$

$$|A| = \frac{1}{210 \cdot 2^{18}}$$

$$|\text{Adj } A^{-1}| = \frac{1}{|A|^2}$$

$$= (210)^2 . 2^{36}$$

$$= (105)^2 . 2^{38}$$

32. The function f(x), that satisfies the condition $f(x) = x + \int_0^{\frac{\pi}{2}} \sin x . \cos y f(y) dy$, is : **[JEE(Main)Aug.2021]**

फलन f(x), जो $f(x) = x + \int_0^{\frac{\pi}{2}} \sin x . \cos y f(y) dy$ को संतुष्ट करता है, है : **[JEE(Main)Aug.2021]**

(A) $x + (\pi - 2) \sin x$ (B) $x + \frac{\pi}{2} \sin x$

(C) $x + \frac{2}{3}(\pi - 2) \sin x$ (D) $x + (\pi + 2) \sin x$

Ans. A

Sol. $f(x) = x + \int_0^{\frac{\pi}{2}} \sin x . \cos y \ f(y) dy$

$$f(x) = x + \sin x \int_0^{\frac{\pi}{2}} \cos y \ f(y) dy$$

Let $A = \int_0^{\frac{\pi}{2}} \cos y . f(y) dy$ _____(I)

$\therefore f(x) = x + A \sin x$ _____(II)

using (I) & (II)

$$A = \int_0^{\frac{\pi}{2}} \cos y (y + A \sin y) dy$$

$$A = \int_0^{\frac{\pi}{2}} y \cos y dy + \frac{A}{2} \int_0^{\frac{\pi}{2}} \sin 2y dy$$

$$A = \left[y\sin y + \cos y\right]_0^{\pi/2} + \frac{A}{2}\left[\frac{-\cos 2y}{2}\right]_0^{\pi/2}$$

$$\Rightarrow A = \left[\frac{\pi}{2} - 1\right] + \frac{A}{4}\left[-1(-1) + 1\right]$$

$$\Rightarrow A = \frac{\pi}{2} - 1 + \frac{A}{2}$$

$A = \pi - 2$

$\therefore\ f(x) = x + (\pi - 2)\sin x$

33. Let [t] denote the greatest integer $\le$ t. Then the value of $8.\int_{-\frac{1}{2}}^{1}\left([2x] + |x|\right)dx$ is _____. :

[JEE(Main)Aug.2021]

माना [t] महत्तम पूर्णांक $\le$ t है। तो $8.\int_{-\frac{1}{2}}^{1}\left([2x] + |x|\right)dx$ का मान है _____ **[JEE(Main)Aug.2021]**

Ans. 5

Sol. $$I = 8\int_{\frac{-1}{2}}^{1}[2x]dx + 8\int_{\frac{-1}{2}}^{1}|x|dx$$

$$I = 8\left(\int_{\frac{-1}{2}}^{0} -1.dx + \int_{0}^{\frac{1}{2}} 0dx + \int_{\frac{1}{2}}^{1} 1dx\right) + 8\left(\int_{\frac{-1}{2}}^{0} -xdx + \int_{0}^{1} xdx\right)$$

$$I = 8\left(\frac{-1}{2} + 0 + \frac{1}{2}\right) + 8\left(\frac{-x^2}{2}\Big|_{\frac{-1}{2}}^{0} + \frac{x^2}{2}\Big|_{0}^{1}\right)$$

$$I = 8\left(\frac{1}{8} + \frac{1}{2}\right) = 5$$

34. Let $g : (0, \infty) \to R$ be a differentiable function such that

$$\int\left(\frac{x(\cos x - \sin x)}{e^x + 1} + \frac{g(x)\left(e^x + 1 - xe^x\right)}{\left(e^x + 1\right)^2}\right)dx = \frac{xg(x)}{e^x + 1} + c\text{, for all } x > 0\text{, where c is an arbitrary constant. Then :}$$

[JEE(Main)June2022]

(A) g is decreasing in $\left(0,\frac{\pi}{4}\right)$ (B) g′ is increasing in $\left(0,\frac{\pi}{4}\right)$

(C) $g+g'$ is increasing in $\left(0,\frac{\pi}{2}\right)$ (D) $g-g'$ is increasing in $\left(0,\frac{\pi}{2}\right)$

माना $g:(0,\infty)\to R$ एक अवकलनीय फलन है जिसके लिए

$\int\left(\frac{x(\cos x-\sin x)}{e^x+1}+\frac{g(x)\left(e^x+1-xe^x\right)}{\left(e^x+1\right)^2}\right)dx=\frac{xg(x)}{e^x+1}+c$, सभी $x>0$ के लिए है, जहाँ c एक स्वेच्छ अचर है। तो :

[JEE(Main)June2022]

(A) $\left(0,\frac{\pi}{4}\right)$ में g ह्रासमान है (B) $\left(0,\frac{\pi}{4}\right)$ में g′ वर्धमान है

(C) $\left(0,\frac{\pi}{2}\right)$ में $g+g'$ वर्धमान है (D) $\left(0,\frac{\pi}{2}\right)$ में $g-g'$ वर्धमान है

Ans. D

Sol. $\int\left(\frac{x(\cos x-\sin x)}{e^x+1}+\frac{g(x)\left(e^x+1-xe^x\right)}{\left(e^x+1\right)^2}\right)dx=\frac{xg(x)}{e^x+1}+c$

On differentiating both sides w.r.t. x, we get

$$\left(\frac{x(\cos x-\sin x)}{e^x+1}+\frac{g(x)\left(e^x+1-xe^x\right)}{\left(e^x+1\right)^2}\right)$$

$$=\frac{\left(e^x+1\right)\left(g(x)+xg'(x)\right)-e^x\cdot x\cdot g(x)}{\left(e^x+1\right)^2}$$

$$\left(e^x+1\right)x(\cos x-\sin x)+g(x)\left(e^x+1-xe^x\right)$$

$$=\left(e^x+1\right)\left(g(x)+xg'(x)\right)-e^x\cdot x\cdot g(x)$$

$$\Rightarrow g'(x)=\cos x-\sin x$$

$$\Rightarrow g(x)=\sin x+\cos x+C$$

g(x) is increasing in $\left(0,\frac{\pi}{4}\right)$

$g''(x)=-\sin x-\cos x<0$

$\Rightarrow$ g'(x) is decreasing function

let $h(x)=g(x)+g'(x)=2\cos x+C$

$\Rightarrow h'(x)=g'(x)+g''(x)=-2\sin x<0$

$\Rightarrow$ h is decreasing

let $\phi(x)=g(x)-g'(x)=2\sin x+C$

$\Rightarrow \phi'(x)=g'(x)-g''(x)=2\cos x>0$

$\Rightarrow$ f is increasing

Hence option (4) is correct.

35. If $\int \frac{1}{x}\sqrt{\frac{1-x}{1+x}}dx=g(x)+c, g(1)=0$, then $g\left(\frac{1}{2}\right)$ is equal to : **[JEE(Main)June2022]**

यदि $\int \frac{1}{x}\sqrt{\frac{1-x}{1+x}}dx=g(x)+c, g(1)=0$ हैं, तो $g\left(\frac{1}{2}\right)$ बराबर है : **[JEE(Main)June2022]**

(A) $\log_e\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)+\frac{\pi}{3}$

(B) $\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)+\frac{\pi}{3}$

(C) $\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)-\frac{\pi}{3}$

(D) $\frac{1}{2}\log_e\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)-\frac{\pi}{6}$

Ans. A

Sol. $\int \frac{1}{x}\sqrt{\frac{1-x}{1+x}}dx=g(x)+c$ $\quad g(1)=0$

$x=\cos\theta$

$dx=-\sin 0\, d\theta$

$\int \frac{1}{\cos\theta}\sqrt{\frac{1-\cos\theta}{1+\sin\theta}}$ $\quad x-\sin\theta\, dx$

$-\int \frac{\sin\theta}{\cos\theta}\sqrt{\frac{\sin^2\left(\frac{\theta}{2}\right)}{\cos^2\left(\frac{\theta}{2}\right)}}d\theta$ $\quad = -\int \frac{\sin\theta}{\cos\theta}\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}d\theta$

$-\int \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\times\sin\frac{\theta}{2}}{\cos\theta\cos\frac{\theta}{2}}d\theta$

$$-\int \frac{2\sin^2\left(\frac{\theta}{2}\right)}{\cos\theta}d\theta = -\int \frac{1-\cos\theta}{\cos\theta}\,d\theta$$

$$-\int \sec\theta d\theta + \int d\theta$$

$$\theta - \int \sec\theta\, d\theta + c$$

$\theta - ln(\sec\theta + \tan\theta) + c$

$\cos^{-1}n - ln(\sec(\cos^{-1}x) + \tan(\cos^{-1}x)) + c$

$g(x) = \cos^{-1}x - ln(\sec(\cos^{-1}x) + \tan^{-1}(\cos^{-1}1))$

$g(1) = 0 - ln(1 + \tan\theta) = 0$

$$(g)\frac{1}{2} = \frac{\pi}{3} - ln\left(\sec\frac{\pi}{3} + \tan\frac{\pi}{3}\right)$$

$$(g)\frac{1}{2} = \frac{\pi}{3} - ln\left(2+\sqrt{3}\right)$$

$$(g)\frac{1}{2} = \frac{\pi}{3} + ln\left(\frac{1}{2+\sqrt{3}}\right) = \frac{\pi}{3} + ln\left(2-\sqrt{3}\right)$$

36. If $\int \frac{\left(x^2+1\right)e^x}{\left(x+1\right)^2}\,dx = f(x)e^x + C$, where C is a constant, then $\frac{d^3f}{dx^3}$ at $x = 1$ is equal to :

[JEE(Main)June2022]

यदि $\int \frac{\left(x^2+1\right)e^x}{\left(x+1\right)^2}\,dx = f(x)e^x + C$ है, जहाँ C एक अचर है, तो $x = 1$ पर $\frac{d^3f}{dx^3}$ बराबर है :

[JEE(Main)June2022]

(A) $-\frac{3}{4}$ (B) $\frac{3}{4}$ (C) $-\frac{3}{2}$ (D) $\frac{3}{2}$

Ans. B

Sol. $$g(x) = \left(\frac{x^2+1}{(x+1)^2}\right)e^x = \left(\frac{x^2+x+1-x}{(x+1)^2}\right)e^x$$

$$g(x) = \left(\frac{x}{x+1} + \frac{1-x}{(x+1)^2}\right)e^x$$

$$g(x) = \left(\frac{x}{x+1} + \frac{1}{(x+1)^2} - \left(\frac{x+1-1}{(x+1)^2}\right)\right)e^x$$

$$g(x)=\left(\frac{x}{x+1}+\frac{1}{(x+1)^2}-\frac{1}{1+x}+\frac{1}{(1+x)^2}\right)e^x dx$$

$$\int g(x).dx = e^x\left(\frac{x}{1+x}-\frac{1}{1+x}\right)+C$$

$$f(x)=\frac{x-1}{1+x}$$

$$f'(x)=\frac{(x+1)-(x-1)}{(x+1)^2}=\frac{2}{(x+1)^2}$$

$$f''(x)=\frac{-4}{(x+1)^3}$$

$$f'''(x)=\frac{12}{(x+1)^4}$$

$$f'''(x)=\frac{12}{16}=\frac{3}{4}$$

37. The value of the integral $\int_{-\pi/2}^{\pi/2}\frac{dx}{\left(1+e^x\right)\left(\sin^6 x+\cos^6 x\right)}$ is equal to : **[JEE(Main)June2022]**

समाकलन $\int_{-\pi/2}^{\pi/2}\frac{dx}{\left(1+e^x\right)\left(\sin^6 x+\cos^6 x\right)}$ का मान बराबर है : **[JEE(Main)June2022]**

(A) 2π (B) 0 (C) π (D) $\frac{\pi}{2}$

Ans. C

Sol. $$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{dx}{\left(1+e^x\right)\left(\sin^6 x+\cos^6 x\right)} \quad \text{————(1)}$$

using $\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$

$$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{dx}{\left(1+e^{-x}\right)\left(\sin^6 x+\cos^6 x\right)}$$

$$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{e^x dx}{\left(e^x+1\right)\left(\sin^6 x+\cos^6 x\right)} \quad \text{————(2)}$$

by (1) + (2)

$$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{\sin^6 x + \cos^6 x}$$

$$2I = 2\int_{0}^{\frac{\pi}{2}} \frac{dx}{\sin^6 x + \cos^6 x}$$

$$I = \int_{0}^{\frac{\pi}{2}} \frac{dx}{1 - 3\sin^2 x \cos^2 x}$$

$$I = \int_{0}^{\frac{\pi}{2}} \frac{dx}{1 - \frac{3}{4}\sin^2 2x}$$

$$I = \int_{0}^{\frac{\pi}{2}} \frac{4dx}{4 - 3\sin^2 2x}$$

$a\int_0^{2a} f(x)dx = 2\int_0^a f(dx)$ If $f(2a - x) = f(x)$

$$I = 2\int_0^{\pi/4} \frac{4.dx}{4 - 3\sin^2 2x}$$

$$I = 2\int_0^{\pi/4} \frac{4\sec^2 2x.dx}{4\sec^2 2x - 3\tan^2 2x}$$

$$I = 2\int_0^{\pi/4} \frac{4\sec^2 2x.dx}{4 + \tan^2 2x}$$

Put $\tan 2x = t$;

$2\sec^2 2x.dx = dt$

$$I = 4\int_0^{\infty} \frac{dt}{4 + t^2}$$

$$I = 4 \times \frac{1}{2}\left[\tan^{-1}\frac{t}{2}\right]_0^{\infty}$$

$$I = 2\left[\tan^{-1}\infty - \tan^{-1}0\right]$$

$$I = 2\left[\frac{\pi}{2} - 0\right]$$

$= \pi$

38. $\lim\limits_{n\to\infty}\left(\frac{n^2}{(n^2+1)(n+1)}+\frac{n^2}{(n^2+4)(n+2)}+....+\frac{n^2}{(n^2+n^2)(n+n)}\right)$ is equal to : **[JEE(Main)June2022]**

$\lim\limits_{n\to\infty}\left(\frac{n^2}{(n^2+1)(n+1)}+\frac{n^2}{(n^2+4)(n+2)}+....+\frac{n^2}{(n^2+n^2)(n+n)}\right)$ बराबर है : **[JEE(Main)June2022]**

(A) $\frac{\pi}{8}+\frac{1}{4}\log_e 2$ (B) $\frac{\pi}{4}+\frac{1}{8}\log_e 2$ (C) $\frac{\pi}{4}-\frac{1}{8}\log_e 2$ (D) $\frac{\pi}{8}+\log_e\sqrt{2}$

Ans. A

Sol. $Tr=\frac{n^2}{(n^2+r^2)(n+r)}$

$$Tr=\frac{\frac{1}{n}}{\left(1+\frac{r^2}{n^2}\right)\left(1+\frac{r}{n}\right)}$$

$$Sum=\sum_{r=1}^{n}\frac{\frac{1}{n}}{\left(1+\frac{r^2}{n^2}\right)\left(1+\frac{r}{n}\right)}$$

$$=\int_0^1\frac{dx}{(1+x^2)(1+x)}$$

$$\frac{1}{(1+x^2)(1+x)}=\frac{A}{1+x}+\frac{Bx+C}{1+x^2}$$

$1 = A(1 + x^2) + (Bx + C)(1 + x)$

Compare

$$A=\frac{1}{2},\ B=-\frac{1}{2},\ C=\frac{1}{2}$$

$$=\int_0^1\left(\frac{\frac{1}{2}}{1+x^2}dx\right)+\int_0^1\frac{\left(-\frac{1}{2}x+\frac{1}{2}\right)}{(1+x^2)}dx$$

$$=\frac{1}{2}\int_0^1\frac{dx}{1+x}-\frac{1}{4}\int_0^1\frac{2xdx}{1+x^2}+\frac{1}{2}\int_0^1\frac{dx}{1+x^2}$$

$$=\frac{1}{2}\ell n(1+x)\Big]_0^1-\frac{1}{4}\ell n(1+x^2)\Big]_0^1+\frac{1}{2}\left[\tan^{-1}x\right]_0^1$$

$$= \frac{1}{2}\ell n2 - \frac{1}{4}\ell n2 + \frac{\pi}{8}$$

$$= \frac{\pi}{8} + \frac{1}{4}\ell n2$$

39. The value of $\int_0^{\pi} \frac{e^{\cos x}\sin x}{\left(1+\cos^2 x\right)\left(e^{\cos x}+e^{-\cos x}\right)}dx$ is equal to : **[JEE(Main)June2022]**

$\int_0^{\pi} \frac{e^{\cos x}\sin x}{\left(1+\cos^2 x\right)\left(e^{\cos x}+e^{-\cos x}\right)}dx$ का मान बराबर है : **[JEE(Main)June2022]**

(A) $\frac{\pi^2}{4}$ (B) $\frac{\pi^2}{2}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Ans. C

Sol. $\int_0^{\pi} \frac{e^{\cos x}\sin x}{\left(1+\cos^2 x\right)\left(e^{\cos x}+e^{-\cos x}\right)}dx$

Use King's property

$$I = \int_0^{\pi} \frac{e^{-\cos x}\sin x}{\left(1+\cos^2 x\right)\left(e^{-\cos x}+e^{\cos x}\right)}dx$$

On adding equation (1) and (2), we get

$$2I = \int_0^{\pi} \frac{\sin x}{1+\cos^2 x}dx = 2\int_0^{\pi/2} \frac{\sin x}{1+\cos^2 x}dx$$

On putting $\cos x = t$, we get

$$I = \int_0^1 \frac{dt}{1+t^2} = \left(\tan^{-1} t\right)_0^1 = \frac{\pi}{4}$$

40. If $b_n = \int_0^{\frac{\pi}{2}} \frac{\cos^2 nx}{\sin x}dx$, $n \in N$, then : **[JEE(Main)June2022]**

(A) $b_3 - b_2$, $b_4 - b_3$, $b_5 - b_4$ are in an A.P. with common difference – 2

(B) $\frac{1}{b_3 - b_2}, \frac{1}{b_4 - b_3}, \frac{1}{b_5 - b_4}$ are in an A.P. with common difference 2

(C) $b_3 - b_2$, $b_4 - b_3$, $b_5 - b_4$ are in a G.P.

(D) $\frac{1}{b_3 - b_2}, \frac{1}{b_4 - b_3}, \frac{1}{b_5 - b_4}$ are in an A.P. with common difference – 2

यदि $b_n = \int_0^{\frac{\pi}{2}} \frac{\cos^2 nx}{\sin x} dx,\ n \in N$, है, तो : **[JEE(Main)June2022]**

(A) $b_3 - b_2, b_4 - b_3, b_5 - b_4$ एक A.P. में हैं, जिसका सार्वअंतर – 2 है

(B) $\frac{1}{b_3 - b_2}, \frac{1}{b_4 - b_3}, \frac{1}{b_5 - b_4}$ एक A.P. में हैं, जिसका सार्वअंतर 2 है

(C) $b_3 - b_2, b_4 - b_3, b_5 - b_4$ एक G.P. में हैं

(D) $\frac{1}{b_3 - b_2}, \frac{1}{b_4 - b_3}, \frac{1}{b_5 - b_4}$ एक A.P. में हैं, जिसका सार्वअंतर – 2 है

Ans. D

Sol. $b_n - b_{n-1} = \int_0^{\frac{\pi}{2}} \frac{\cos^2 nx - \cos^2 (n-1)x}{\sin x} dx$

$= \int_0^{\frac{\pi}{2}} \frac{\sin(2n-1)x \sin(-x)}{\sin x} dx$

$= \left[\frac{\cos(2n-1)x}{2n-1} \right]_0^{\frac{\pi}{2}}$

$b_n - b_{n-1} = \frac{-1}{2n-1}$

$\frac{1}{b_n - b_{n-1}} = -2n+1$, A. P. with common difference "–2".

41. The value of the integral $\int_{-2}^{2} \frac{|x^3 + x|}{\left(e^{x|x|} + 1\right)} dx$ is equal to : **[JEE(Main)June2022]**

समाकलन $\int_{-2}^{2} \frac{|x^3 + x|}{\left(e^{x|x|} + 1\right)} dx$ का मान बराबर है : **[JEE(Main)June2022]**

(A) $5e^2$ (B) $3e^{-2}$ (C) 4 (D) 6

Ans. D

Sol. $I = \int_{-2}^{2} \frac{|x^3 + x|}{\left(e^{x|x|} + 1\right)} dx$ ——(1)

apply king rule

$I = \int_{-2}^{2} \frac{|-x^3 - x|}{e^{-x|x|} + 1} dx$

$$I=\int_{-2}^{2}\frac{\left|x^3+x\right|}{e^{-x|x|}+1}dx \qquad \text{———(2)}$$

(1) + (2)

$$2I=\int_{-2}^{2}\left|x^3+x\right|\left(e^{\frac{1}{x|x|+1}}+\frac{1}{e^{-x|x|}+1}\right)dx$$

$$2I=\int_{-2}^{2}\left|x^3+x\right|\left(\frac{e^{x|x|}+1}{e^{x|x|}+1}\right)dx$$

$$2I=\int_{-2}^{2}\left|x^3+x\right|dx$$

$$2I=2\int_{0}^{2}\left|x^3+x\right|dx$$

$$I=\int_{0}^{2}\left(x^3+x\right)dx$$

$$I=\left(\frac{x^4}{4}+\frac{x^2}{2}\right)_0^2=4+2=6$$

42. If m and n respectively are the number of local maximum and local minimum points of the function $f(x)=\int_0^{x^2}\frac{t^2-5t+4}{2+e^t}dt$, then the ordered pair (m, n) is equal to : **[JEE(Main)June2022]**

यदि फलन $f(x)=\int_0^{x^2}\frac{t^2-5t+4}{2+e^t}dt$ के स्थानीय उच्चतम तथा स्थानीय निम्नतम बिन्दुओं की संख्या क्रमशः m तथा n है, तो क्रमित युग्म (m, n) बराबर है : **[JEE(Main)June2022]**

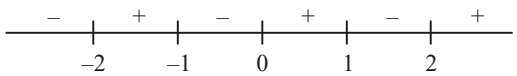
(A) (3, 2) (B) (2, 3) (C) (2, 2) (D) (3, 4)

Ans. B

Sol. $f'(x)=\frac{x^4-5x^2+4}{2+e^{x^2}}.2x$

$$=\frac{2x\left(x^2-1\right)\left(x^2-4\right)}{e^{x^2}+2}$$

$$=\frac{2(x-1)(x-2)x(x+1)(x+2)}{e^{x^2}+2}$$



Minima at x = – 2, 0, 2

maxima at x = – 1, 1

m = 2, n = 3 (2, 3)

43. Let f be a differentiable function in $\left(0,\frac{\pi}{2}\right)$. If $\int_{\cos x}^{1} t^2 f(t)dt = \sin^3 x + \cos x$ then $\frac{1}{\sqrt{3}} f'\left(\frac{1}{\sqrt{3}}\right)$ is equal to :

[JEE(Main)June2022]

माना $\left(0,\frac{\pi}{2}\right)$ में f एक अवकलनीय फलन है। यदि $\int_{\cos x}^{1} t^2 f(t)dt = \sin^3 x + \cos x$ है, तो $\frac{1}{\sqrt{3}} f'\left(\frac{1}{\sqrt{3}}\right)$ बराबर है :

[JEE(Main)June2022]

(A) $6-9\sqrt{2}$ (B) $6-\frac{9}{\sqrt{2}}$ (C) $\frac{9}{2}-6\sqrt{2}$ (4) $\frac{9}{\sqrt{2}}-6$

Ans. B

Sol. $\int_{\cos x}^{1} t^2 f(t)\,dt = \sin^2 x + \cos x$

Differentiate both side

$-\cos^2 x\, f(\cos x)(-\sin x) = 3\sin^2 x \cos x - \sin x$

$\Rightarrow f(\cos x) = \frac{(3\sin x \cos x - 1)}{\cos^2 x}$

$\Rightarrow$ f(cos x) = 3 tan x – sec^{2}x

Again differentiating wrt 'x'

$f'(\cos x)(-\sin x) = 3\sec^2 x - 2\sec^2 x \tan x$

put $\cos x = \frac{1}{\sqrt{3}}$ $\Rightarrow \sin x = \frac{\sqrt{2}}{\sqrt{3}}$

$\frac{-\sqrt{2}}{\sqrt{3}} f\left(\frac{1}{\sqrt{3}}\right) = 3.(3) - 2(3)\sqrt{2}$

$\frac{1}{\sqrt{3}} f\left(\frac{1}{\sqrt{3}}\right) = \frac{3\left(3-2\sqrt{2}\right)}{-\sqrt{2}}$

$= 6 - \frac{9}{\sqrt{2}}$

44. The integral $\int_{0}^{1} \frac{1}{7^{\left\lfloor \frac{1}{x} \right\rfloor}} dx$, where [.] denotes the greatest integer function, is equal to :

[JEE(Main)June2022]

समाकलन $\int_0^1 \frac{1}{7^{\left[\frac{1}{x}\right]}}dx$, जहाँ [.] महत्तम पूर्णांक फलन है, बराबर है : **[JEE(Main)June2022]**

(A) $1+6\ \log_e\left(\frac{6}{7}\right)$ (B) $1-6\log_e\left(\frac{6}{7}\right)$

(C) $\log_e\left(\frac{7}{6}\right)$ (D) $1-7\log_e\left(\frac{6}{7}\right)$

Ans. A

Sol. $\int_0^1 \frac{1}{7^{\left[\frac{1}{x}\right]}}dx$

$=-\int_0^1 \frac{dx}{7^{\left[\frac{1}{x}\right]}}$

$-\left[\int_1^{\frac{1}{2}} \frac{1}{7^1}dx+\int_{\frac{1}{2}}^{\frac{1}{3}} \frac{1}{7^2}dx+\int_{\frac{1}{3}}^{\frac{1}{4}} \frac{1}{7^3}dx+.....\right]$

$-\left[\frac{1}{7}\left(\frac{1}{2}-1\right)+\frac{1}{7^2}\left(\frac{1}{3}-\frac{1}{2}\right)+\frac{1}{7^3}\left(\frac{1}{4}-\frac{1}{3}\right)+.....\right]$

$\left(\frac{1}{7}+\frac{1}{2.7^2}+\frac{1}{3.7^3}+.....\right)-\left(\frac{1}{7.2}+\frac{1}{7^2.3}+\frac{1}{7^3.4}+.....\right)$

$-\ell n\left(1-\frac{1}{7}\right)+7\left(\ell n\left(1-\frac{1}{7}\right)-\frac{1}{7}\right)$

$=6\ \ell n\frac{6}{7}+1$

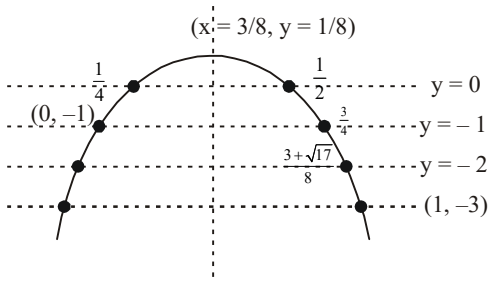
45. Let [t] denote the greatest integer less than or equal to t. Then, the value of the integral $\int_0^1\left[-8x^2+6x-1\right]dx$ is equal to : **[JEE(Main)June2022]**

माना [t] महत्तम पूर्णांक ≤ t है। तो समाकलन $\int_0^1\left[-8x^2+6x-1\right]dx$ का मान बराबर है : **[JEE(Main)June2022]**

(A) -1 (B) $\frac{-5}{4}$ (C) $\frac{\sqrt{17}-13}{8}$ (D) $\frac{\sqrt{17}-16}{8}$

Ans. C

Sol.



Let $y = -8x^2 + 6x - 1$

for $y = -2$

$-8x^2 + 6x - 1 = 0 - 2$

$\Rightarrow x = \dfrac{3+\sqrt{17}}{8}$

for $y = -1$

$-8x^2 + 6x - 1 = -1$

$\Rightarrow x = \dfrac{3}{4}$

$$\text{so } \int_0^1 \left[-8x^2 + 6x - 1\right]dx = \int_0^{1/4} (-1)\,dx + \int_{1/4}^{1/2} 0\,dx + \int_{1/2}^{3/4} (-1)\,dx + \int_{3/4}^{\frac{3+\sqrt{17}}{8}} (-2)\,dx + \int_{\frac{3+\sqrt{17}}{8}}^{1} (-3)\,dx$$

$$-\frac{1}{4}\times 1 + 0 + -1\left(\frac{3}{4} - \frac{1}{2}\right) - \left(\frac{3+\sqrt{17}}{8} - \frac{3}{4}\right)(2) - \left(1 - \frac{3+\sqrt{17}}{8}\right)3$$

$$= \frac{\sqrt{17} - 13}{8}$$

46. Let $f : R \to R$ be a differentiable function such that $f\left(\dfrac{\pi}{4}\right) = \sqrt{2}$, $f\left(\dfrac{\pi}{2}\right) = 0$ and $f'\left(\dfrac{\pi}{2}\right) = 1$ and let $g(x) = \int_x^{\frac{\pi}{4}} \left(f'(t)\sec t + \tan t \sec t\, f(t)\right)dt$ for $x \in \left[\dfrac{\pi}{4}, \dfrac{\pi}{2}\right)$. Then $\lim\limits_{x \to \left(\frac{\pi}{2}\right)^-} g(x)$ is equal to :

[JEE(Main)June2022]

माना $f : R \to R$ एक अवकलनीय फलन है जिसके लिए $f\left(\dfrac{\pi}{4}\right) = \sqrt{2}$, $f\left(\dfrac{\pi}{2}\right) = 0$ तथा $f'\left(\dfrac{\pi}{2}\right) = 1$ है। माना $g(x) = \int_x^{\frac{\pi}{4}} \left(f'(t)\sec t + \tan t \sec t\, f(t)\right)dt$, $x \in \left[\dfrac{\pi}{4}, \dfrac{\pi}{2}\right)$ है। तो $\lim\limits_{x \to \left(\frac{\pi}{2}\right)^-} g(x)$ बराबर है :

[JEE(Main)June2022]

(A) 2 (B) 3 (C) 4 (D) –3

Ans. B

Sol. Given $f\left(\frac{\pi}{4}\right)=\sqrt{2}$, $f\left(\frac{\pi}{2}\right)=0$ and $f'\left(\frac{\pi}{2}\right)=1$

Also

$$g(x)=\int_x^{\frac{\pi}{4}}\left[f'(t).\sec t+\tan t.\sec t.f(t)\right]dt$$

$$\because \int(u).v+v'.u).dx=\int(u.v).dx=u.v.$$

$$\therefore g(x)=\left[\sec t.f(t)\right]_x^{\frac{\pi}{4}}$$

$g(x) = 2 - \sec x.f(x)$

Now,

$$\lim_{x\to\frac{\pi}{2}} g(x)=\lim_{h\to 0} g\left(\frac{\pi}{2}-h\right)=2-\lim_{h\to 0}\sec\left(\frac{\pi}{2}-h\right).f\left(\frac{\pi}{2}-h\right)$$

$$\lim_{x\to\frac{\pi}{2}} g(x)=2-\lim_{h\to 0}\operatorname{cosec} h\, f\left(\frac{\pi}{2}-h\right)$$

$$=2-\lim_{h\to 0}\frac{f\left(\frac{\pi}{2}-h\right)}{\cos h}$$

using Lopital rules

$$=2+\lim_{h\to 0}\frac{f'\left(\frac{\pi}{2}-h\right)}{\cos h}=2+1=3$$

47. Let $f: R \to R$ be a continuous function satisfying $f(x) + f(x + k) = n$, for all $x \in R$ where $k > 0$ and n is a positive integer. If $I_1=\int_0^{4nk} f(x)dx$ and $I_2=\int_{-k}^{3k} f(x)dx$, then : **[JEE(Main)June2022]**

माना $f: R \to R$ एक संतत फलन है जो सभी $x \in R$ के लिए $f(x) + f(x + k) = n$ को संतुष्ट करता है, जहाँ $x > 0$ है तथा n एक धनात्मक पूर्णांक है। यदि $I_1=\int_0^{4nk} f(x)dx$ तथा $I_2=\int_{-k}^{3k} f(x)dx$ हैं, तो : **[JEE(Main)June2022]**

(A) $I_1 + 2I_2 = 4nk$ (B) $I_1 + 2I_2 = 2nk$ (C) $I_1 + nI_2 = 4n^2k$ (D) $I_1 + nI_2 = 6n^2k$

Ans. C

Sol. Given $f: R \to R$ and $f(x) + f(x + k) = n \ \forall \ x \in R$ ————(1)

for $x \to x + k$

$f(x + k) + f(x + 2k) = n$ ————(2)

so $f(x + 2k) = f(x)$ [from (1) and (2)]

therefore period of $f(x)$ is $2k$.

Now,

$$I_1 = \int_0^{4nk} f(x).dx = 2n \int_0^{2K} f(x).dx$$

$$= 2n\left[\int_0^{k} f(x).dx + \int_k^{2k} f(x).dx\right]$$

in $\int_k^{2k} f(x).dx$ put $x = t + k$ $\Rightarrow dx = dt$

$$= 2n\left[\int_0^{k} f(x).dx + \int_0^{k} f(t+k).dt\right] = 2n.(nk)$$

$=2n^2k$

Also

in $I_2 = \int_{-k}^{3k} f(x).dx = 2\int_0^{2k} f(x).dx = 2nk$

$\Rightarrow I_1 + I_2 \times n = 4n^2k$

48. Let $f : R \to R$ be a function by :

माना एक फलन $f : R \to R$

$$f(x) = \begin{cases} \max\limits_{t \le x}\{t^3 - 3t\} & ; \quad x \le 2 \\ x^2 + 2x - 6 & ; \quad 2 < x < 3 \\ [x-3] + 9 & ; \quad 3 \le x \le 5 \\ 2x + 1 & ; \quad x > 5 \end{cases}$$

where [t] is the greatest integer less than or equal to t. Let m be the number of points where f is not differentiable and $I = \int_{-2}^{2} f(x)\,dx$. Then the ordered pair (m, I) is equal to : **[JEE(Main)June2022]**

द्वारा परिभाषित हैं, जहाँ [t] महत्तम पूर्णांक $\le t$ है। माना m उन बिन्दुओं की संख्या हैं, जहाँ f अवकलनीय नहीं हैं तथा $I = \int_{-2}^{2} f(x)\,dx$ है। तो क्रमित युग्म (m, I) बराबर है : **[JEE(Main)June2022]**

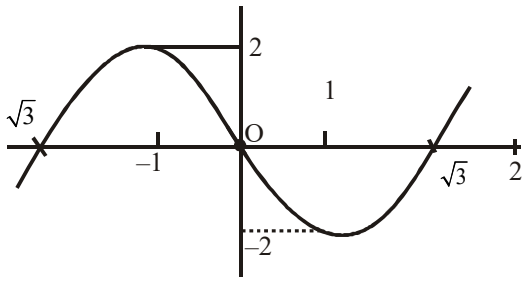
(A) $\left(3, \frac{27}{4}\right)$ (B) $\left(3, \frac{23}{4}\right)$ (C) $\left(4, \frac{27}{4}\right)$ (D) $\left(4, \frac{23}{4}\right)$

Ans. C

Sol. $\max \{t^3 - 3t\}; x \le 2$

$t \le x$

$g(t) = t^3 - 3t \Rightarrow g'(t) = 3(t^2 - 1)$



clearly $f(x)\begin{cases} x^3 - 3x & x < -1 \\ 2 & -1 \le x \le 2 \\ x^2 + 2x - 6 & 2 < x < 3 \\ 9 & 3 \le x < 4 \\ 10 & 4 \le x < 5 \\ 11 & x = 5 \\ 2x + 1 & x > 5 \end{cases}$

Non diff. at $x = 2, 3, 4, 5$

$m = 4$

$$I = \int_{-2}^{2} f(x)dx = \int_{-2}^{-1} \left(x^3 - 3x\right)dx + \int_{-1}^{2} 2dx$$

$$= \frac{27}{4}$$

Ans. $\left(4, \frac{27}{4}\right)$

49. $\int_0^5 \cos\left(\pi\left(x - \left[\frac{x}{2}\right]\right)\right)dx$ where [t] denotes greatest integer less than or equal to t, is equal to :

[JEE(Main)June2022]

$\int_0^5 \cos\left(\pi\left(x - \left[\frac{x}{2}\right]\right)\right)dx$ जहाँ [t] महत्तम पूर्णांक $\le t$ है, बराबर है : **[JEE(Main)June2022]**

(A) -3 (B) -2 (C) 2 (D) 0

Ans. D

Sol. $\int_0^5 \cos\left(\pi\left(x-\left[\frac{x}{2}\right]\right)\right)dx = \int_0^2 \cos\left(\pi\left(x-\left[\frac{x}{2}\right]\right)\right)dx$

$$+\int_2^4 \cos\left(\pi\left(x-\left[\frac{x}{2}\right]\right)\right)dx$$

$$+\int_4^5 \cos\left(\pi\left(-.\left[\frac{x}{2}\right]\right)\right)dx$$

$$=\frac{\sin\pi x}{\pi}\Big)_0^2+\frac{\sin\pi(x-1)}{\pi}\Big)_2^4+\frac{\sin\pi(x-2)}{\pi}\Big)_4^5$$

$= 0$

50. Let f be a real valued continuous function on [0, 1] and $f(x) = x + \int_0^1 (x-t)f(t)\,dt$. Then, which of the following points (x, y) lies on the curve y = f(x)? **[JEE(Main)June2022]**

माना [0, 1] पर f एक वास्तविक मान संतत फलन है तथा $f(x) = x + \int_0^1 (x-t)f(t)\,dt$ है। तो निम्न में से कौनसा बिन्दु (x, y) वक्र y = f(x) पर स्थित है ? **[JEE(Main)June2022]**

(A) (2, 4) (B) (1, 2) (C) (4, 17) (D) (6, 8)

Ans. D

Sol. Given that $f(x) = x\int_0^1 (x-t)f(t)\,dt$

$$f(x) = x + \int_0^1 f(t)\,dt - \int_0^1 tf(t)\,dt$$

$$f(x) = x\left[tf\int_0^1 f(t)\,dt\right] - \int_0^1 tf(t)\,dt$$

Let $1+\int_0^1 f(t)\,dt = a$ and $\int_0^1 tf(t)\,dt = 1$

$\therefore$ f(x) = a × –b

So, $a = 1+\int_0^1 (at-b)dt = 1+\frac{a}{2}-b \quad \Rightarrow \frac{a}{2}+b=1$ _____(1)

$b = \int_0^t t(at-b)\,dt = \frac{a}{3}-\frac{b}{2} \quad \Rightarrow \frac{3b}{2}=\frac{a}{3}$ _____(2)

Solve (1) and (2)

$a = \frac{18}{13},\ b = \frac{4}{13}$

$\therefore f(x) = = \frac{18\times(-4)}{13}$

So, (6, 8) lies on f(10).

51. If $\int_0^2\left(\sqrt{2x}-\sqrt{2x-x^2}\right)dx = \int_0^1\left(1-\sqrt{1-y^2}-\frac{y^2}{2}\right)dy + \int_1^2\left(2-\frac{y^2}{2}\right)dy + I$ then I equals : **[JEE(Main)June2022]**

यदि $\int_0^2\left(\sqrt{2x}-\sqrt{2x-x^2}\right)dx = \int_0^1\left(1-\sqrt{1-y^2}-\frac{y^2}{2}\right)dy + \int_1^2\left(2-\frac{y^2}{2}\right)dy + I$ है, तो I बराबर है:

[JEE(Main)June2022]

(A) $\int_0^1\left(1+\sqrt{1-y^2}\right)dy$

(B) $\int_0^1\left(\frac{y^2}{2}-\sqrt{1-y^2}+1\right)dy$

(C) $\int_0^1\left(1-\sqrt{1-y^2}\right)dy$

(D) $\int_0^1\left(\frac{y^2}{2}+\sqrt{1-y^2}+1\right)dy$

Ans. C

Sol. Given $\int_0^2\sqrt{2x}\,dx - \int_0^2\sqrt{1-(x-1)^2}\,dx$

$= \int_0^2\left(1-\frac{y^2}{2}\right)dy - \int_0^1\sqrt{1-y^2}\,dy + 1 + I$

$\Rightarrow \frac{8}{3} - 2\int_0^1\sqrt{1-y^2}dy = \frac{2}{3} + 1 - \int_0^1\sqrt{1-y^2}dy + I$

$\Rightarrow I = 1 - \int_0^1\sqrt{1-y^2}dy$

$\Rightarrow I = \int_0^1\left(1-\sqrt{1-y^2}dy\right)$

52. Let $f(\theta) = \sin\theta + \int_{-\pi/2}^{\pi/2}(\sin\theta + t\cos\theta)f(t)dt$. Then value of $\left|\int_0^{\pi/2} f(\theta)d\theta\right|$ is _____.

[JEE(Main)June2022]

माना $f(\theta) = \sin\theta + \int_{-\pi/2}^{\pi/2}(\sin\theta + t\cos\theta)f(t)dt$ है। तो $\left|\int_0^{\pi/2} f(\theta)d\theta\right|$ का मान है _____

[JEE(Main)June2022]

Ans. 1

Sol. $f(\theta)=\sin\theta+\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(\sin\theta+t\cos\theta)f(t)dt$

$$f(\theta)=\sin\theta+\sin\theta+\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}f(t)dt+\cos\theta\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}}tf(t)dt$$

Let $A=\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}}f(t)dt, B=\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}}tf(t)dt$

$$f(\theta)=\sin\theta+A\sin\theta+B\cos\theta$$

$$f(\theta)=(A+1)\sin\theta+B\cos\theta$$

$$A=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(A+1)\sin t+B\cos t\,dt$$

$A=2B$ ———(1)

$$B=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}t\left((A+1)\sin t+B\cos t\right)$$

$$B=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}t(A+1)\sin t$$

$$B=(A+1)2\int_{0}^{\frac{\pi}{2}}t\sin t\,dt$$

$B = (A + 1)\ 2.1$

$2A + 2 – B = 0$ ———(2)

After solving

$$B=-\frac{2}{3}, A=-\frac{4}{3}$$

$$\left|\int_{0}^{\frac{\pi}{2}}f(\theta)d\theta\right|=\left|\int_{0}^{\frac{\pi}{2}}-\frac{1}{3}\sin\theta-\frac{2}{3}\cos\theta\right|$$

$= 1$

53. Let $\underset{0\le x\le 2}{\text{Max}}\left\{\frac{9-x^2}{5-x}\right\}=\alpha$ and $\underset{0\le x\le 2}{\text{Min}}\left\{\frac{9-x^2}{5-x}\right\}=\beta$. If $\int\limits_{\beta-\frac{8}{3}}^{2\alpha-1}\text{Max}\left\{\frac{9-x^2}{5-x},x\right\}dx=\alpha_1+\alpha_2\log_e\left(\frac{8}{15}\right)$ then

$\alpha_1+\alpha_2$ is equal to _____. **[JEE(Main)June2022]**

माना $\underset{0\le x\le 2}{\text{Max}}\left\{\frac{9-x^2}{5-x}\right\}=\alpha$ तथा $\underset{0\le x\le 2}{\text{Min}}\left\{\frac{9-x^2}{5-x}\right\}=\beta$ हैं। यदि $\int\limits_{\beta-\frac{8}{3}}^{2\alpha-1}\text{Max}\left\{\frac{9-x^2}{5-x},x\right\}dx=\alpha_1+\alpha_2\log_e\left(\frac{8}{15}\right)$ है, तो

$\alpha_1+\alpha_2$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 34

Sol. Let $f(x)=\frac{x^2-9}{x-5}\Rightarrow f'(x)=\frac{(x-1)(x-9)}{(x-5)^2}$

So, $\alpha=f(1)=2$ and $\beta=\min(f(0),f(2))=\frac{5}{3}$

Now, $\int\limits_{-1}^{3}\max\left\{\frac{x^2-9}{x-5},x\right\}dx=\int\limits_{-1}^{\frac{9}{5}}\frac{x^2-9}{x-5}dx+\int\limits_{\frac{9}{5}}^{3}xdx$

$=\int\limits_{-1}^{\frac{9}{5}}\left(x+5+\frac{16}{x-5}\right)dx+\frac{x^2}{2}\Big|_{\frac{9}{5}}^{3}$

$=\frac{28}{25}+14+16\ln\left(\frac{8}{15}\right)+\frac{72}{25}=18+16\ln\left(\frac{8}{15}\right)$

Clearly $\alpha_1=18$ and $\alpha_2=16$, so $\alpha_1+\alpha_2=34$

54. The value of $b>3$ for which $12\times\int\limits_3^b\frac{1}{(x^2-1)(x^2-4)}dx=\log_e\left(\frac{49}{40}\right)$, is equal to _____.

[JEE(Main)June2022]

$b>3$ का वह मान, जिसके लिए $12\times\int\limits_3^b\frac{1}{(x^2-1)(x^2-4)}dx=\log_e\left(\frac{49}{40}\right)$ है, है _____

[JEE(Main)June2022]

Ans. 6

Sol. $12.\int_3^b\frac{1}{(x^2-1)(x^2-4)}dx=\ln\left(\frac{49}{40}\right)$

$12.\int_3^b\left(\frac{-1}{3}.\frac{1}{x^2-1}+\frac{1}{3}.\frac{1}{x^2-4}\right)dx=\ln\left(\frac{49}{40}\right)$

$$\left[-4.\frac{1}{2}\ln\left(\frac{x-1}{x+1}\right)+4.\frac{1}{4}\ln\left(\frac{x-2}{x+2}\right)\right]_3^b=\ln\left(\frac{49}{40}\right)$$

$$\left[\ln\left\{\frac{x-2}{x+2}.\left(\frac{x+1}{x-1}\right)^2\right\}\right]_3^b=\ln\left(\frac{49}{50}\right)$$

$$\ln\left\{\frac{b-2}{b+2}.\left(\frac{b+1}{b-1}\right)^2\right\}-\ln\left\{\frac{1}{5}.4\right\}=\ln\left(\frac{49}{40}\right)$$

$$\ln\left\{\frac{b-2}{b+2}.\left(\frac{b+1}{b-1}\right)^2\right\}=\ln\left\{\frac{49}{40}.\frac{4}{5}\right\}$$

$$\ln\left\{\frac{b-2}{b+2}.\left(\frac{b+1}{b-1}\right)^2\right\}=\ln\left\{\frac{4}{8}.\left(\frac{7}{5}\right)^2\right\}$$

b = 6

55. The value of the integral $\frac{48}{\pi^4}\int_0^{\pi}\left(\frac{3\pi x^2}{2}-x^3\right)\frac{\sin x}{1+\cos^2 x}dx$ is equal to _____.

[JEE(Main)June2022]

समाकलन $\frac{48}{\pi^4}\int_0^{\pi}\left(\frac{3\pi x^2}{2}-x^3\right)\frac{\sin x}{1+\cos^2 x}dx$ का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. 6

Sol. $$I=\frac{48}{\pi^4}\int_0^{\pi}x^2\left(\frac{3\pi}{2}-x\right)\frac{\sin x}{1+\cos^2 x}dx \quad ...(1)$$

Apply king property

$$I=\frac{48}{\pi^4}\int_0^{\pi}(\pi-x)^2\left(\frac{\pi}{2}+x\right)\frac{\sin x}{1+\cos^2 x}dx \quad ...(2)$$

(A) + (2)

$$I=\frac{12}{\pi}\int_0^{\pi}\frac{\sin x dx}{1+\cos^2 x}$$

Let $\cos x = t \Rightarrow \sin x\,dx = -dt$

$$I=\frac{12}{\pi}\int_1^{-1}\frac{-dt}{1+t^2}=6$$

56. The integral $\frac{24}{\pi}\int_0^{\sqrt{2}} \frac{\left(2-x^2\right)dx}{\left(2+x^2\right)\sqrt{4+x^4}}$ is equal to _____. **[JEE(Main)June2022]**

समाकलन $\frac{24}{\pi}\int_0^{\sqrt{2}} \frac{\left(2-x^2\right)dx}{\left(2+x^2\right)\sqrt{4+x^4}}$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 3

Sol. $$=\frac{24}{\pi}\int_0^{\sqrt{2}} \frac{x^2\left(\frac{2}{x^2}-1\right)dx}{x\left(\frac{2}{x}+x\right)x\sqrt{\frac{4}{x^2}+x^2}}$$

$$=\frac{24}{\pi}\int_0^{\sqrt{2}} \frac{\left(\frac{2}{x^2}-1\right)dx}{\left(\frac{2}{x}+x\right)\sqrt{\left(\frac{2}{x}+x\right)^2-4}}$$

Let $t=\frac{2}{x}+x$

$$-dt=\left(\frac{2}{x^2}-1\right)dx$$

$$=-\frac{24}{\pi}\int\frac{dt}{t\sqrt{t^2-4}}$$

$$=-\frac{24}{\pi}\times\frac{1}{2}\left[\sec^{-1}\left(\frac{t}{2}\right)\right]$$

$$=-\frac{12}{\pi}\left[\sec^{-1}\left(\frac{x}{2}+\frac{1}{x}\right)\right]_0^{\sqrt{2}}$$

$$=-\frac{12}{\pi}\left[\sec^{-1}\sqrt{2}-\sec^{-1}\infty\right]$$

$$=-\frac{12}{\pi}\left[\frac{\pi}{4}-\frac{\pi}{2}\right]$$

$$=3$$

57. The integral $\int \frac{\left(1-\frac{1}{\sqrt{3}}\right)(\cos x-\sin x)}{\left(1+\frac{2}{\sqrt{3}}\sin 2x\right)}dx$ is equal to : **[JEE(Main)July2022]**

समाकलन $\int \frac{\left(1-\frac{1}{\sqrt{3}}\right)(\cos x-\sin x)}{\left(1+\frac{2}{\sqrt{3}}\sin 2x\right)}dx$ बराबर है : **[JEE(Main)July2022]4**

(A) $\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}\right|+C$

(B) $\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{3}\right)}\right|+C$

(C) $\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}\right|+C$

(D) $\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}-\frac{\pi}{6}\right)}\right|+C$

Ans. A

Sol. $I=\int \frac{\left(1-\frac{1}{\sqrt{3}}\right)(\cos x-\sin x)}{\left(1+\frac{2}{\sqrt{3}}\sin 2x\right)}dx$

$\frac{\sqrt{3}}{2}=\int \frac{\left(1-\frac{1}{\sqrt{3}}\right)(\cos x-\sin x)}{\left(\frac{\sqrt{3}}{2}+\sin 2x\right)}dx$

$\int \frac{\left(\frac{\sqrt{3}}{2}-\frac{1}{2}\right)(\cos x-\sin x)}{\sin 60^o+\sin 2x}dx$

$\int \frac{\left(\frac{\sqrt{3}}{2}\cos x-\frac{1}{2}\cos x-\frac{\sqrt{3}}{2}\sin x+\frac{1}{2}\sin x\right)}{2\sin\left(x+\frac{\pi}{6}\right)\cos\left(x-\frac{\pi}{6}\right)}dx$

$$\int \frac{\left(\cos\left(x-\frac{\pi}{6}\right)-\sin\left(x+\frac{\pi}{6}\right)\right)}{2\sin\left(x+\frac{\pi}{6}\right)\cos\left(x-\frac{\pi}{6}\right)}dx$$

$$\frac{1}{2}\left(\int \frac{dx}{\sin\left(x+\frac{\pi}{6}\right)}-\int \frac{dx}{\cos\left(x-\frac{\pi}{6}\right)}\right)$$

$$\frac{1}{2}\ln\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}\right|$$

58. For $I(x)=\int \frac{\sec^2 x-2022}{\sin^{2022} x}dx$, if $I\left(\frac{\pi}{4}\right)=2^{1011}$, then : **[JEE(Main)July2022]**

$I(x)=\int \frac{\sec^2 x-2022}{\sin^{2022} x}dx$ के लिए, यदि $I\left(\frac{\pi}{4}\right)=2^{1011}$ है, तो : **[JEE(Main)July2022]**

(A) $3^{1010} I\left(\frac{\pi}{3}\right)-I\left(\frac{\pi}{6}\right)=0$ (B) $3^{1010} I\left(\frac{\pi}{6}\right)-I\left(\frac{\pi}{3}\right)=0$

(C) $3^{1011} I\left(\frac{\pi}{3}\right)-I\left(\frac{\pi}{6}\right)=0$ (D) $3^{1011} I\left(\frac{\pi}{6}\right)-I\left(\frac{\pi}{3}\right)=0$

Ans. A

Sol. $I(x)=\int \sec^2 x\cdot \sin^{-2022} x dx-2022\int \sin^{-2022} x dx$

$=\tan x\cdot(\sin x)^{-2022}+\int (2022)\tan x\cdot(\sin x)^{-2023}\cos x dx$

$-2022\int (\sin x)^{-2022}dx$

$I(x)=(\tan x)(\sin x)^{-2022}+C$

At $X=\pi/4, 2^{1011}=\left(\frac{1}{\sqrt{2}}\right)^{-2022}+C \therefore C=0$

Hence $I(x)=\frac{\tan x}{(\sin x)^{2022}}$

$$I(\pi/6)=\frac{1}{\sqrt{3}\left(\frac{1}{2}\right)^{2022}}=\frac{2^{2022}}{\sqrt{3}}$$

$$I(\pi/3)=\frac{\sqrt{3}}{\left(\frac{\sqrt{3}}{2}\right)^{2022}}=\frac{2^{2022}}{(\sqrt{3})^{2021}}=\frac{1}{3^{1010}}I\left(\frac{\pi}{6}\right)$$

$$3^{1010}I(\pi/3)=I(\pi/6)$$

59. For any real number x, let[x] denote the largest integer less than equal to x. Let f be a real valued function defined on the interval [–10, 10] by $f(x)=\begin{cases} x-[x], & \text{if } [x] \text{is odd} \\ 1+[x]-x, & \text{if } [x] \text{is even} \end{cases}$. Then the value of $\frac{\pi^2}{10}\int_{-10}^{10} f(x)\cos\pi x\,dx$ is :

[JEE(Main)July2022]

किसी वास्तविक संख्या x के लिए, माना [x] महत्तम पूर्णांक ≤ x है। माना अंतराल [–10, 10] में एक वास्तविक मान फलन $f(x)=\begin{cases} x-[x], & \text{यदि } [x] \text{विषम है} \\ 1+[x]-x, & \text{यदि } [x] \text{सम है} \end{cases}$ द्वारा परिभाषित है। तो $\frac{\pi^2}{10}\int_{-10}^{10} f(x)\cos\pi x\,dx$ का मान है :

[JEE(Main)July2022]

(A) 4 (B) 2 (C) 1 (D) 0

Ans. A

Sol. $f(x)=\begin{cases} \{x\}, & 2n-1\le x<2n \\ 1-\{x\}, & 2n\le x<2n+1 \end{cases}$

Clearly f(x) is a periodic function with period = 2

Hence f(x) . cos πx is also periodic with period = 2

$$\frac{\pi^2}{10}\int_{-10}^{10} f(x)\cos(\pi x)\,dx=\pi^2\int_0^2 f(x)\cos(\pi x)\,dx=\pi^2\int_0^1((1-\{x\})+\{-x\})\cos(\pi x)\,dx$$

$$=2\pi^2\int_0^1(-x\cos\pi x)dx=-2\pi^2\left[\frac{x\sin\pi x}{\pi}+\frac{\cos\pi x}{\pi^2}\right]_0^1=-2\pi^2\left(-\frac{2}{\pi^2}\right)=4$$

60. The slope of the tangent to a curve C : y = y(x) at any point (x, y) on it is $\frac{2e^{2x}-6e^{-x}+9}{2+9e^{-2x}}$. If C passes through the points $\left(0,\frac{1}{2}+\frac{\pi}{2\sqrt{2}}\right)$ and $\left(\alpha,\frac{1}{2}e^{2\alpha}\right)$, then e^{α} is equal to : **[JEE(Main)July2022]**

वक्र C : y = y(x) के किसी बिन्दु (x, y) पर स्पर्श रेखा की प्रवणता $\frac{2e^{2x}-6e^{-x}+9}{2+9e^{-2x}}$ है। यदि C, बिन्दुओं $\left(0,\frac{1}{2}+\frac{\pi}{2\sqrt{2}}\right)$ तथा $\left(\alpha,\frac{1}{2}e^{2\alpha}\right)$ से होकर जाता है, तो e^{α} बराबर है : **[JEE(Main)July2022]**

(A) $\frac{3+\sqrt{2}}{3-\sqrt{2}}$ (B) $\frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$ (C) $\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$ (D) $\frac{\sqrt{2}+1}{\sqrt{2}-1}$

Ans. B

Sol. $\frac{dy}{dx}=\frac{2e^{2x}-3e^{-x}+9}{2+9e^{-2x}}$

$\frac{dy}{dx}=e^{2x}-\frac{6e^{x}}{2e^{2x}+9}$

$y=\frac{e^{2x}}{2}-\tan^{-1}\left(\frac{\sqrt{2}e^{x}}{3}\right)+c$

If C passes through the point $\left(0,\frac{1}{2}+\frac{\pi}{2\sqrt{2}}\right)$

$c=-\frac{\pi}{4}-\tan^{-1}\frac{\sqrt{2}}{3}$

Again C passes through the point $\left(\alpha,\frac{1}{2}e^{2\alpha}\right)$

then $e^{\alpha}=\frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$

61. $\lim\limits_{n\to\infty}\frac{1}{2^n}\left(\frac{1}{\sqrt{1-\frac{1}{2^n}}}+\frac{1}{\sqrt{1-\frac{2}{2^n}}}+\frac{1}{\sqrt{1-\frac{3}{2^n}}}+.....+\frac{1}{\sqrt{1-\frac{2^n-1}{2^n}}}\right)$ is equal to : **[JEE(Main)July2022]**

$\lim\limits_{n\to\infty}\frac{1}{2^n}\left(\frac{1}{\sqrt{1-\frac{1}{2^n}}}+\frac{1}{\sqrt{1-\frac{2}{2^n}}}+\frac{1}{\sqrt{1-\frac{3}{2^n}}}+.....+\frac{1}{\sqrt{1-\frac{2^n-1}{2^n}}}\right)$ बराबर है : **[JEE(Main)July2022]**

(A) $\frac{1}{2}$ (B) 1 (C) 2 (D) –2

Ans. C

Sol. $I=\lim\limits_{n\to\infty}\dfrac{1}{2^n}\left(\dfrac{1}{\sqrt{1-\dfrac{1}{2^n}}}+\dfrac{1}{\sqrt{1-\dfrac{2}{2^n}}}+\dfrac{1}{\sqrt{1-\dfrac{3}{2^n}}}+.....+\dfrac{1}{\sqrt{1-\dfrac{2^n-1}{2^n}}}\right)$

Let $2^n=t$ and if $n\to\infty$ then $t\to\infty$

$$I=\lim_{n\to\infty}\frac{1}{t}\left(\sum_{r=1}^{t-1}\frac{1}{\sqrt{1-\dfrac{r}{t}}}\right)$$

$$I=\int_0^1\frac{dx}{\sqrt{1-x}}=\int_0^1\frac{dx}{\sqrt{x}}\quad\left(\int_0^0 f(x)dx=\int_0^0 f(a-x)dx\right.$$

$$=\left[2x^{\frac{1}{2}}\right]_0^1=2$$

62. Let [t] denote the greatest integer less than or equal to t. Then the value of the integral $\int_{-3}^{101}\left(\left[\sin\left(\pi x\right)\right]+e^{\left[\cos(2\pi x)\right]}\right)dx$ is equal to : **[JEE(Main)July2022]**

माना कि [t] वो महत्तम पूर्णांक है जो t से कम या बराबर है। तो समाकलन $\int_{-3}^{101}\left(\left[\sin\left(\pi x\right)\right]+e^{\left[\cos(2\pi x)\right]}\right)dx$ का मान बराबर है: **[JEE(Main)July2022]**

(A) $\dfrac{52(1-e)}{e}$ (B) $\dfrac{52}{e}$ (C) $\dfrac{52(2+e)}{e}$ (D) $\dfrac{104}{e}$

Ans. B

Sol. $I=\int_{-3}^{101}\left([\sin(\pi x)]+e^{[\cos(2\pi x)]}\right)dx$

$[\sin\pi x]$ is periodic with period 2 and $e^{[\cos(2\pi x)]}$ is periodic with period 1.

So,

$$I=52\int_0^2\left([\sin\pi x]+e^{[\cos 2\pi x]}\right)dx$$

$$=52\left\{\int_1^2-1dx+\int_{\frac{1}{4}}^{\frac{3}{4}}e^{-1}dx+\int_{\frac{5}{4}}^{\frac{7}{4}}e^{-1}dx+\int_0^{\frac{1}{4}}e^0dx+\int_{\frac{3}{4}}^{\frac{5}{4}}e^0dx+\int_{\frac{7}{4}}^2e^0dx\right\}$$

$$=\frac{52}{e}$$

63. If $a=\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\frac{2n}{n^2+k^2}$ and $f(x)=\sqrt{\frac{1-\cos x}{1+\cos x}}$, $x\in(0,1)$, then : **[JEE(Main)July2022]**

यदि $a=\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\frac{2n}{n^2+k^2}$ तथा $f(x)=\sqrt{\frac{1-\cos x}{1+\cos x}}$, $x\in(0,1)$ हैं, तो : **[JEE(Main)July2022]**

(A) $2\sqrt{2}f\left(\frac{a}{2}\right)=f'\left(\frac{a}{2}\right)$ (B) $f\left(\frac{a}{2}\right)f'\left(\frac{a}{2}\right)=\sqrt{2}$

(C) $\sqrt{2}f\left(\frac{a}{2}\right)=f'\left(\frac{a}{2}\right)$ (D) $f\left(\frac{a}{2}\right)=\sqrt{2}f'\left(\frac{a}{2}\right)$

Ans. C

Sol. $a=\frac{1}{n}\sum\limits_{k=1}^{n}\frac{2}{1+\left(\frac{k}{n}\right)^2}=\int_0^1\frac{2}{1+x^2}dx=\frac{\pi}{2}$

$f(x)=\tan\left(\frac{x}{2}\right);x\in(0,1)$

$f\left(\frac{\pi}{4}\right)=\sqrt{2}-1$

$f'\left(\frac{\pi}{4}\right)=\frac{1}{2}\sec^2\left(\frac{\pi}{8}\right)=\frac{\sqrt{2}}{\sqrt{2}+1}$

$f'\left(\frac{\pi}{4}\right)=\sqrt{2}f\left(\frac{\pi}{4}\right)$

64. $\int\limits_0^{20\pi}\left(|\sin x|+|\cos x|\right)^2dx$ is equal to : **[JEE(Main)July2022]**

$\int\limits_0^{20\pi}\left(|\sin x|+|\cos x|\right)^2dx$ बराबर है : **[JEE(Main)July2022]**

(A) $10(\pi+4)$ (B) $10(\pi+2)$ (C) $20(\pi-2)$ (D) $20(\pi+2)$

Ans. D

Sol. $I=\int\limits_0^{20\pi}\left(|\sin x|+|\cos x|\right)^2dx$; (Jack property)

$I=40\int\limits_0^{\pi/2}\left(\sin x+\cos x\right)^2dx$

$I = 40\int_0^{\pi/2}(1+\sin 2x)\,dx$

$I = 20[\pi+2]$

65. Let $f: R \to R$ be a function defined as $f(x) = a\sin\left(\frac{\pi[x]}{2}\right)+[2-x]$, $a \in R$, where [t] is the greatest integer less than or equal to t if $\lim_{x\to -1} f(x)$ exists, then the value of $\int_0^4 f(x)\,dx$ is equal to : **[JEE(Main)July2022]**

माना एक फलन $f: R \to R$, $f(x) = a\sin\left(\frac{\pi[x]}{2}\right)+[2-x]$, $a \in R$, द्वारा परिभाषित है, जहाँ [t] महत्तम पूर्णांक $\le t$ है। यदि $\lim_{x\to -1} f(x)$ का अस्तित्व है, तो $\int_0^4 f(x)\,dx$ का मान बराबर है : **[JEE(Main)July2022]**

(A) –1 (B) –2 (C) 1 (D) 2

Ans. B

Sol. $\lim_{x\to -1^+} a\sin\left(\pi\frac{[x]}{2}\right)+[2-x] = -a+2$

$\lim_{x\to -1^-} a\sin\left(\pi\frac{[x]}{2}\right)+[2-x] = 0+3 = 3$

$\lim_{x\to -1^-} f(x)$ exist when $a = -1$

Now,

$\int_0^4 f(x)\,dx = \int_0^1 f(x)\,dx + \int_1^2 f(x)\,dx + \int_2^3 f(x)\,dx + \int_3^4 f(x)\,dx$

$= \int_0^1 (0+1)\,dx + \int_1^2 (-1+0)\,dx + \int_2^3 (0-1)\,dx + \int_3^4 (1-2)\,dx$

$= 1 - 1 - 1 - 1 = -2$

66. Let $I = \int_{\pi/4}^{\pi/3}\left(\frac{8\sin x - \sin 2x}{x}\right)dx$. Then : **[JEE(Main)July2022]**

माना $I = \int_{\pi/4}^{\pi/3}\left(\frac{8\sin x - \sin 2x}{x}\right)dx$ है, तो : **[JEE(Main)July2022]**

(A) $\frac{\pi}{2} < I < \frac{3\pi}{4}$ (B) $\frac{\pi}{4} < I < \frac{5\pi}{12}$ (C) $\frac{5\pi}{12} < I < \frac{\sqrt{2}}{3}\pi$ (D) $\frac{3\pi}{4} < I < \pi$

Ans. C

Sol. Consider

$f(x) = 8\sin x - \sin 2x$

$f'(x) = 8\sin x - 2\cos 2x$

$f''(x) = -8\sin x + 4\sin 2x$

$= -8\sin x(1 - \cos x)$

$\therefore f'(x) < 0\, x \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$

$\therefore f'(x)$ is $\downarrow$ function

$f'\left(\frac{\pi}{3}\right) < f'(x) < f'\left(\frac{\pi}{4}\right)$

$5 < f'(x) < \frac{8}{\sqrt{2}}$

$5 < f'(x) < 4\sqrt{2}$

$5x < f(x) < 4\sqrt{2}x$

$5 < \frac{f(x)}{x} < 4\sqrt{2}$

$\int_{\pi/4}^{\pi/3} 5 < \int \frac{f(x)}{x} < \int_{\pi/4}^{\pi/3} 4\sqrt{2}$

$\int_{\pi/4}^{\pi/3} 5 < \int \frac{8\sin x - \sin 2x}{x} < \int_{\pi/4}^{\pi/3} 4\sqrt{2}$

$\frac{5\pi}{12} < I < \frac{\sqrt{2}\pi}{3}$

67. Let $f(x) = 2 + |x| - |x - 1| + |x + 1|$, $x \in R$. Consider

(S1) : $f'\left(-\frac{3}{2}\right) + f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) = 2$

(S2) : $\int_{-2}^{2} f(x)\,dx = 12$

Then : **[JEE(Main)July2022]**

(A) Both (S1) and (S2) are correct (B) Both (S1) and (S2) are wrong

(C) Only (S1) is correct (D) Only (S2) is correct

माना $f(x) = 2 + |x| - |x-1| + |x+1|, x \in R$ है। तो

$(S1): f'\left(-\frac{3}{2}\right)+f'\left(-\frac{1}{2}\right)+f'\left(\frac{1}{2}\right)+f'\left(\frac{3}{2}\right)=2$

$(S2): \int_{-2}^{2} f(x)\,dx = 12$

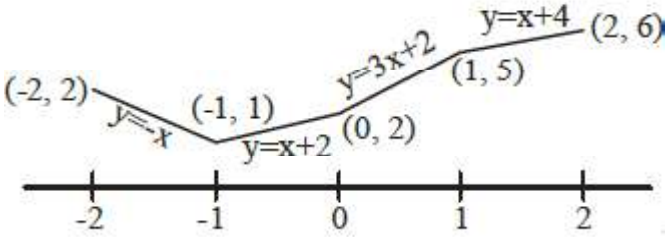
में :

(A) (S1) तथा (S2) दोनों सही हैं (B) (S1) तथा (S2) दोनों गलत हैं

(C) केवल (S1) सही है (D) केवल (S2) सही है

Ans. D

Sol.



$(S1): f'\left(-\frac{3}{2}\right)+f'\left(-\frac{1}{2}\right)+f'\left(\frac{1}{2}\right)+f'\left(\frac{3}{2}\right)=4$

$(S2): \int_{-2}^{2} f(x)\,dx = 12$

$\therefore$ (4)

68. $\int_0^2 \left(\left|2x^2-3x\right|+\left[x-\frac{1}{2}\right]\right)dx$, where [t] is the greatest integer function, is equal to :

[JEE(Main)July2022]

$\int_0^2 \left(\left|2x^2-3x\right|+\left[x-\frac{1}{2}\right]\right)dx$, जहाँ [t] महत्तम पूर्णांक फलन है, बराबर है : **[JEE(Main)July2022]**

(A) $\frac{7}{6}$ (B) $\frac{19}{12}$ (C) $\frac{31}{12}$ (D) $\frac{3}{2}$

Ans. B

Sol. $\int_0^2 \left|2x^2-3x\right|dx$

$$= \int_0^{\frac{3}{2}} \left(3x-2x^2\right)dx + \int_{\frac{3}{2}}^{2} \left(2x^2-3x\right)dx = \frac{19}{12}.$$

$$\int_0^2 \left[x - \frac{1}{2} \right] dx = \int_{\frac{-1}{2}}^{\frac{3}{2}} [t] dt$$

$$= \int_{\frac{1}{2}}^{0} (-1)\, dt + \int_0^1 0.dt \int_1^{\frac{3}{2}} 1.dt = 0.$$

69. Let $I_n(x) = \int_0^x \frac{1}{(t^2+5)^n} dt$, n = 1, 2, 3, ... Then : **[JEE(Main)July2022]**

माना $I_n(x) = \int_0^x \frac{1}{(t^2+5)^n} dt$, n = 1, 2, 3, ... है। तो : **[JEE(Main)July2022]**

(A) $50I_6 - 9I_5 = xI_5'$ (B) $50I_6 - 11I_5 = xI_5'$

(C) $50I_6 - 9I_5 = I_5'$ (D) $50I_6 - 11I_5 = I_5'$

Ans. A

Sol. $I_n(x) = \int_0^x \frac{dt}{(t^2+5)^n}$

Applying integral by parts

$$I_n(x) = \left[\frac{t}{(t^2+5)^n} \right]_0^x - \int_0^x n(t^2+5)^{-n-1}.2t^2$$

$$I_n(x) = \frac{x}{(x^2+5)^n} + 2n \int_0^x \frac{t^2}{(t^2+5)^{n+1}} dt$$

$$I_n(x) = \frac{x}{(x^2+5)^n} + 2n \int_0^x \frac{(t^2+5)-5}{(t^2+5)^{n+1}} dt$$

$$I_n(x) = \frac{x}{(x^2+5)^n} + 2n\, I_n(x) - 10n\, I_{n+1}(x)$$

$$10n\, I_{n+1}(x) + (1-2n) I_n(x) = \frac{x}{(x^2+5)^n}$$

Put n = 5

70. The integral $\int_0^{\frac{\pi}{2}} \frac{1}{3+2\sin x+\cos x}dx$ is equal to : **[JEE(Main)July2022]**

समाकलन $\int_0^{\frac{\pi}{2}} \frac{1}{3+2\sin x+\cos x}dx$ बराबर है : **[JEE(Main)July2022]**

(A) $\tan^{-1}(2)$ (B) $\tan^{-1}(2)-\frac{\pi}{4}$

(C) $\frac{1}{2}\tan^{-1}(2)-\frac{\pi}{8}$ (D) $\frac{1}{2}$

Ans. B

Sol. $I=\int_0^{\frac{\pi}{2}} \frac{dx}{3+2\sin x+\cos x}=\int_0^{\frac{\pi}{2}} \frac{\sec^2\frac{x}{2}.dx}{2\tan^2\frac{x}{2}+4\tan\frac{x}{2}+4}$

Put $\tan\frac{x}{2}=t,$ so

$I=\int_0^1 \frac{dt}{(t+1)^2+1}=\tan^{-1}(x+1)\Big|_0^1=\tan^{-1}2-\frac{\pi}{4}$

71. If $f(\alpha)=\int_1^{\alpha} \frac{\log_{10}t}{1+t}dt$, $\alpha>0$, then $f(e^3)+f(e^{-3})$ is equal to : **[JEE(Main)July2022]**

यदि $f(\alpha)=\int_1^{\alpha} \frac{\log_{10}t}{1+t}dt$, $\alpha>0$ है, तो $f(e^3)+f(e^{-3})$ बराबर है : **[JEE(Main)July2022]**

(A) 9 (B) $\frac{9}{2}$ (C) $\frac{9}{\log_e(10)}$ (D) $\frac{9}{2\log_e(10)}$

Ans. D

Sol. $f(e^3)=\int_1^{e^3} \frac{\ell nt}{\ell n10(1+t)}dt$ .....(1)

$f(\alpha)=\int_1^{\alpha} \frac{\ell nt}{(\ell n10)(1+t)}dt$

$t=\frac{1}{x}\Rightarrow x=\frac{1}{t}$

$$dt = \frac{-1}{x^2}dx$$

$$= \int_1^{\frac{1}{\alpha}} \frac{-\ell nx}{(\ell n10)\left(1+\frac{1}{x}\right)}\left(-\frac{1}{x^2}\right)dx$$

$$f(\alpha) = \frac{1}{\ell n10}\int_1^{\frac{1}{\alpha}} \frac{\ell nx}{x(x+1)}dx$$

$$f\left(e^{-3}\right) = \frac{1}{\ell n10}\int_1^{e^3} \frac{\ell nt}{t(t+1)}dt \text{ .....(2)}$$

Add (1) & (2)

$$\left(e^3\right) + f\left(e^{-3}\right)$$

$$= \left(\frac{1}{\ell n10}\right)\int_1^{e^3} \frac{\ell nt}{(1+t)}\left[1+\frac{1}{t}\right]dt$$

$$= \left(\frac{1}{\ell n10}\right)\int_1^3 \frac{\ln t}{t}dt$$

$$\ell nt = r$$

$$\frac{dt}{t} = dr$$

$$= \frac{1}{\ell n10}\int_0^3 rdr$$

$$= \left(\frac{1}{\ell n10}\right)\left(\frac{r^2}{2}\right)\Bigg|_0^3$$

$$= \left(\frac{1}{\log 10}\right)\left(\frac{9}{2}\right)$$

$$= \frac{9}{2\log_e 10}$$

72. If [t] denotes the greatest integer $\le t$, then the value of $\int_0^1 \left[2x - \left|3x^2 - 5x + 2\right| + 1\right]dx$ is :

[JEE(Main)July2022]

यदि [t] महत्तम पूर्णांक $\le t$ है, तो $\int_0^1 \left[2x - \left|3x^2 - 5x + 2\right| + 1\right]dx$ का मान है : **[JEE(Main)July2022]**

(A) $\frac{\sqrt{37}+\sqrt{13}-4}{6}$ (B) $\frac{\sqrt{37}-\sqrt{13}-4}{6}$

(C) $\frac{-\sqrt{37}-\sqrt{13}+4}{6}$ (D) $\frac{-\sqrt{37}+\sqrt{13}+4}{6}$

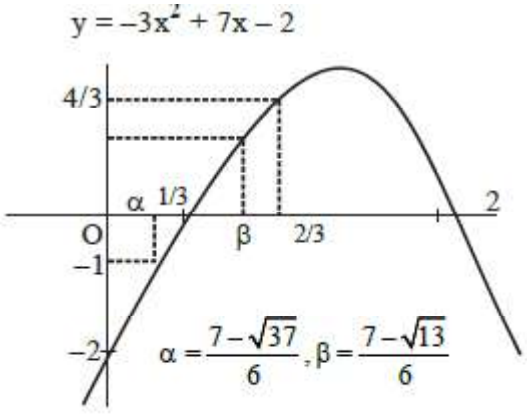
Ans. A

Sol. $I=\int_0^1\left[2x-\left|3x^2-3x-2x+2I+1\right|\right]dx$

$I=\int_0^1\left[2x-\left|(3x-2)(x-1)\right|\right]dx+\int_0^1 1dx$

$I=\int_0^{2/3}\left[\left(2x-\left(3x^2-5x+2\right)\right)\right]dx+\int_{2/3}^1\left(2x+\left(3x^2-5x+2\right)\right)dx+1$

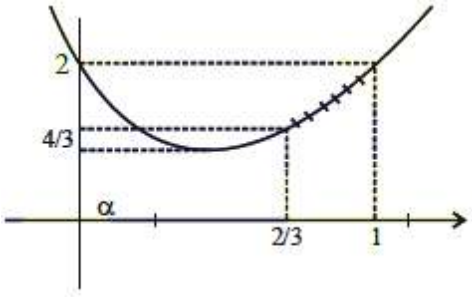
$I=\int_0^{2/3}\left[-3x^2+7x-2\right]dx+\int_{2/3}^1\left(3x^2-3x+2\right)dx+1$



$\int_0^{\alpha}(-2)dx+\int_{\alpha}^{1/3}(-1)dx+\int_{1/3}^{\beta}0dx+\int_{\beta}^{2/3}1\cdot dx$

$=-2\alpha-\left(\frac{1}{3}-\alpha\right)+\frac{2}{3}-\beta=-\alpha-\beta+\frac{1}{3}$

$y=3x^2-3x+2$



When $x\in\left(\frac{2}{3},1\right)$

$3x^2-3x+2\in\left(\frac{4}{3},2\right)$

$\left[3x^2-3x+2\right]=1$

$\therefore \int_{2/3}^{1}\left[3x^2-3x+2\right]dx=1\left(1-\frac{2}{3}\right)=\frac{1}{3}$

Hence $I=\left(\frac{1}{3}-(\alpha+\beta)\right)+\left(\frac{1}{3}\right)+1$

$=\frac{5}{3}-\left(\frac{7-\sqrt{37}}{6}+\frac{7-\sqrt{13}}{6}\right)$

$=\frac{-2}{3}+\frac{\sqrt{37}+\sqrt{13}}{6}$

$=\frac{\sqrt{37}+\sqrt{13}-4}{6}$

73. If $\lim\limits_{n\to\infty}\frac{(n+1)^{k-1}}{n^{k+1}}\left[(nk+1)+(nk+2)+...+(nk+n)\right]=33.\ \lim\limits_{n\to\infty}\frac{1}{n^{k+1}}.\left[1^k+2^k+3^k+...+n^k\right]$, then the integral value of k is equal to _____. **[JEE(Main)July2022]**

यदि $\lim\limits_{n\to\infty}\frac{(n+1)^{k-1}}{n^{k+1}}\left[(nk+1)+(nk+2)+...+(nk+n)\right]=33.\ \lim\limits_{n\to\infty}\frac{1}{n^{k+1}}.\left[1^k+2^k+3^k+...+n^k\right]$ हैं, तो k का पूर्णांक मान बराबर है _____ **[JEE(Main)July2022]**

Ans. 5

Sol. LHS

$\lim\limits_{n\to\infty}\frac{(n+1)^{k-1}}{n^{k+1}}[nk.n+1+2+....+n]$

$=\lim\limits_{n\to\infty}\frac{(n+1)^{k-1}}{n^{k+1}}.\left[n^2k+\frac{n(n+1)}{2}\right]$

$=\lim\limits_{n\to\infty}\frac{(n+1)^{k-1}.n^2\left(k+\frac{\left(1+\frac{1}{n}\right)}{2}\right)}{n^{k+1}}$

$$\Rightarrow \lim_{n\to\infty}\left(1+\frac{1}{n}\right)\left(\frac{k+\left(1+\frac{1}{n}\right)}{2}\right)$$

$$\Rightarrow \left(k+\frac{1}{2}\right)$$

RHS

$$\Rightarrow \lim_{n\to\infty}\frac{1}{n^{k+1}}\left(1^k+2^k+....+n^k\right)=\frac{1}{k+1}$$

LHS = RHS

$$\Rightarrow k+\frac{1}{2}=33.\frac{1}{k+1}$$

$\Rightarrow (2k + 1)(k + 1) = 66$

$\Rightarrow (k - 5)(2k + 13) = 0$

$$\Rightarrow k = 5 \text{ or } -\frac{13}{2}$$

74. Let f be a twice differentiable function on R. If $f'(0) = 4$ and

$f(x)+\int_0^x (x-t)f'(t)dt=\left(e^{2x}+e^{-2x}\right)\cos 2x+\frac{2}{a}x$, then $(2a + 1)^5 a^2$ is equal to _____.

[JEE(Main)July2022]

माना R पर दो बार अवकलनीय एक फलन f है। यदि $f'(0) = 4$ है तथा

$f(x)+\int_0^x (x-t)f'(t)dt=\left(e^{2x}+e^{-2x}\right)\cos 2x+\frac{2}{a}x$ है, तो $(2a + 1)^5 a^2$ बराबर है _____

[JEE(Main)July2022]

Ans. 8

Sol. $\because \quad f(x)+\int_0^x (x-t)f'(t)dt=\left(e^{2x}+e^{-2x}\right)$

$$\cos 2x+\frac{2x}{a} \qquad .....(i)$$

Here $f(0) = 2$ .....(ii)

On differentiating equation (i) w.r.t. x we get :

$$f'(x)+\int_0^x f'(t)dt+xf'(x)-xf'(x)=2\left(e^{2x}-e^{-2x}\right)$$

$$\cos 2x - 2\left(e^{2x} + e^{-2x}\right)\sin 2x + \frac{2}{a}$$

$$\Rightarrow f'(x) + f(x) - f(0) = 2\left(e^{2x} - e^{-2x}\right)\cos 2x - 2\left(e^{2x} + e^{-2x}\right)$$

$$\sin 2x + \frac{2}{a}$$

Replace x by 0 we get

$$\Rightarrow 4 = \frac{2}{a} \quad \Rightarrow a = \frac{1}{2}.$$

$$\therefore \left(2a+1\right)^5 .a^2 = 2^5 . \frac{1}{2^2} = 2^3 = 8$$

75. Let $a_n = \int_{-1}^{n}\left(1 + \frac{x}{2} + \frac{x^2}{3} + ... + \frac{x^{n-1}}{n}\right)dx$ for every $n \in N$. Then the sum of all the elements of the set $\{n \in N : a_n \in (2, 30)\}$ is _____. **[JEE(Main)July2022]**

माना प्रत्येक $n \in N$ के लिए $a_n = \int_{-1}^{n}\left(1 + \frac{x}{2} + \frac{x^2}{3} + ... + \frac{x^{n-1}}{n}\right)dx$ है। तो समुच्चय $\{n \in N : a_n \in (2, 30)\}$ के सभी अवयवों का योगफल है _____ **[JEE(Main)July2022]**

Ans. 5

Sol. $\because a_n = \int_{-1}^{n}\left(1 + \frac{x}{2} + \frac{x^2}{3} + ... + \frac{x^{n-1}}{n}\right)dx$

$$= \left[x + \frac{x^2}{2^2} + \frac{x^3}{3^2} + ... + \frac{x^n}{n^2}\right]_{-1}^{n}$$

$$a_n = \frac{n+1}{1^2} + \frac{n^2-1}{2^2} + \frac{n^3+1}{3^2} + \frac{n^4-1}{4^2} + ... + \frac{n^n + (-1)^{n+1}}{n^2}$$

Here $a_1 = 2,\ a_2 = \frac{2+1}{1} + \frac{2^2-1}{2} = 3 + \frac{3}{2} = \frac{9}{2}$

$$a_3 = 4 + 2 + \frac{28}{9} = \frac{100}{9}$$

$$a_4 = 5 + \frac{15}{4} + \frac{65}{9} + \frac{255}{16} > 31$$

$\therefore$ The required set is $\{2, 3\}$. $\because a_n \in (2, 30)$

$\therefore$ Sum of elements = 5.

76. If $n(2n+1)\int_0^1 (1-x^n)^{2n}\,dx = 1177\int_0^1 (1-x^n)^{2n+1}\,dx$, then $n \in N$ is equal to _____.

[JEE(Main)July2022]

यदि $n(2n+1)\int_0^1 (1-x^n)^{2n}\,dx = 1177\int_0^1 (1-x^n)^{2n+1}\,dx$ है, तो $n \in N$ बराबर है_____

[JEE(Main)July2022]

Ans. Official Answer NTA (24)

Sol. Let $I_1 = \int_0^1 (1-x^n)^{2n}\,dx, I_2 = \int_0^1 (1-x^n)^{2n+1}\,dx$

$I_2 = \int_0^1 (1-x^n)^{2n+1}.1\,dx$

$= (1-x^n)^{2n+1}.x\,|_0^1 - \int_0^1 (2n+1)(1-x^n)^{2n}(-nx^{n-1})x\,dx$

$I_2 = -n(2n+1)\{I_2 + I_1\}$

$(2n^2+n+1)I_2 = n(2n+1)I_1$

$\frac{I_1}{I_2} = \frac{2n^2+n+1}{n(2n+1)} = \frac{1177}{n(2n+1)}$

$\Rightarrow 2n^2+n-1176 = 0 \Rightarrow n = 24$

77. Let f(x) = min{[x – 1], [x – 2], ..., [x – 10]} where [t] denotes the greatest integer ≤ t. Then $\int_0^{10} f(x)\,dx + \int_0^{10} (f(x))^2\,dx + \int_0^{10} |f(x)|\,dx$ is equal to : _____. **[JEE(Main)July2022]**

माना f(x) = min{[x – 1], [x – 2], ..., [x – 10]} है, जहाँ [t] महत्तम पूर्णांक ≤ t है। तो

$\int_0^{10} f(x)\,dx + \int_0^{10} (f(x))^2\,dx + \int_0^{10} |f(x)|\,dx$ बराबर है _____ **[JEE(Main)July2022]**

Ans. 385

Sol. f(x) = [x] – 10

$\int_0^{10} f(x).dx = -10-9-8-.....-1$

$= -\frac{10.11}{2} = -55$

$\int_0^{10} (f(x))^2\,dx = 10^2+9^2+8^2+....+1^2$

$$= \frac{10.11.21}{6} = 385$$

$$\int_0^{10} |f(x)| = 10 + 9 + 8 + ... + 1$$

$$= \frac{10.11}{2} = 55$$

$$= -55 + 385 + 55 = 385$$

78. Let f be differentiable function satisfying $f(x) = \frac{2}{\sqrt{3}} \int_0^{\sqrt{3}} f\left(\frac{\lambda^2 x}{3}\right) d\lambda$, $x > 0$ and $f(1) = \sqrt{3}$. If $y = f(x)$ passes through the point $(\alpha, 6)$ then α is equal to _____. **[JEE(Main)July2022]**

माना f एक अवकलनीय फलन है, $f(x) = \frac{2}{\sqrt{3}} \int_0^{\sqrt{3}} f\left(\frac{\lambda^2 x}{3}\right) d\lambda$, $x > 0$ तथा $f(1) = \sqrt{3}$ हैं। यदि $y = f(x)$ बिन्दु $(\alpha, 6)$ से होकर जाता है, तो α बराबर है _____ **[JEE(Main)July2022]**

Ans. 12

Sol. Let, $\frac{\lambda^2 x}{3} = t$

$$\Rightarrow \frac{2\lambda x}{3} d\lambda = dt$$

$$\Rightarrow d\lambda = \frac{3}{2} . \frac{1\sqrt{x}}{x.\sqrt{3}\sqrt{t}} dt$$

$$\Rightarrow d\lambda = \frac{\sqrt{3}}{2} . \frac{1}{\sqrt{x}} . \frac{dt}{\sqrt{t}}$$

So, $f(x) = \frac{1}{\sqrt{x}} \int_0^x \frac{f(t)}{\sqrt{t}} dt$

$$\Rightarrow \sqrt{x}.f'(x) + \frac{f(x)}{2\sqrt{x}} = \frac{f(x)}{\sqrt{x}}$$

$$\Rightarrow \sqrt{x}.f'(x) = \frac{f(x)}{2\sqrt{x}}$$

$$\Rightarrow \frac{dy}{y} = \frac{dx}{2x}$$

$\Rightarrow \ln y = \frac{1}{2}\ln x + c \Rightarrow f(x) = \sqrt{x}$

$\Rightarrow y = \sqrt{3x} \quad \{\text{as } f(1) = \sqrt{3}\}$

So, $f(x) = \sqrt{3x}$

Now, $f(\alpha) = 6 \Rightarrow 36 = 3\alpha$

$\Rightarrow \alpha = 12$

79. If $\int_0^{\sqrt{3}} \frac{15x^3}{\sqrt{1+x^2+\sqrt{(1+x^2)^3}}}dx = \alpha\sqrt{2} + \beta\sqrt{3}$, where α, β are integers, then $\alpha + \beta$ is equal to _____.

[JEE(Main)July2022]

यदि $\int_0^{\sqrt{3}} \frac{15x^3}{\sqrt{1+x^2+\sqrt{(1+x^2)^3}}}dx = \alpha\sqrt{2} + \beta\sqrt{3}$ है, जहाँ α, β पूर्णांक है, तो $\alpha + \beta$ बराबर_____

[JEE(Main)July2022]

Ans. 10

Sol. Put $1 + x^2 = t^2$

$2x\,dx = 2t\,dt$

$X\,dx = t\,dt$

$\therefore \int_1^2 \frac{15(t^2-1)t\,dt}{\sqrt{t^2+t^3}}$

$15\int_1^2 \frac{t(t^2-1)}{t\sqrt{1+t}}dt$

Put $1 + t = u^2$

$dt = 2u\,du$

$15\int_{\sqrt{2}}^{\sqrt{3}} (u^4 - 2u^2)du$

$30\left(\frac{u^5}{5} - \frac{2u^3}{3}\right)_{\sqrt{2}}^{\sqrt{3}}$

$30\left[\frac{1}{5}\left(\sqrt{3}^5 - \sqrt{2}^5\right) - \frac{2}{5}\left(\sqrt{3}^3 - \sqrt{2}^3\right)\right]$

$$30\left[\frac{1}{5}\left(9\sqrt{3}-4\sqrt{2}\right)-\frac{2}{3}\left(3\sqrt{3}-2\sqrt{2}\right)\right]$$

$$30\left[-\frac{1}{5}+\sqrt{3}+\frac{8}{15}\sqrt{2}\right]$$

$$-6\sqrt{3}+16\sqrt{2}=\alpha\sqrt{2}+\beta\sqrt{3}$$

$$\alpha=16,\beta=-6$$

$$\therefore \alpha+\beta=10$$

80. The value of the integral $\int_0^{\frac{\pi}{2}} 60\frac{\sin(6x)}{\sin x}dx$ is equal to _____. **[JEE(Main)July2022]**

समाकलन $\int_0^{\frac{\pi}{2}} 60\frac{\sin(6x)}{\sin x}dx$ का मान बराबर है _____ **[JEE(Main)July2022]**

Ans. 104

Sol. $$I=60\int_0^{\pi/2}\left(\frac{\sin 6x-\sin 4x}{\sin x}+\frac{\sin 4x-\sin 2x}{\sin x}+\frac{\sin 2x}{\sin x}\right)dx$$

$$I=60\int_0^{\pi/2}\left(2\cos 5x+2\cos 3x+2\cos x\right)dx$$

$$I=60\left(\frac{2}{5}\sin 5x+\frac{2}{3}\sin 3x+2\sin x\right)\Big|_0^{\pi/2}=104$$